Role Differentiation as Ignition of a Collective Information Engine: Structuration in Agent Populations

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Informational active matter research has shown how measurement-informed decisions produce collective order, so far in systems that reach consensus. We identify a collective information engine that orders by differentiation instead, and construct a minimal instance using anti-coordination games, the simplest game class in which differentiated role information has value. Agents infer their role from a noisy social signal grounded in a persistent identity, and role-following action feeds back into that signal, which shapes the incentive to follow roles. Resources accrued through coordinated role-play combine with identity variability to reinforce the schemas that generated them. The model thereby operationalizes Sewell's duality of schemas and resources in structuration, the qualitative resolution of sociology's structure–agency debate. The engine ignites when a loop gain Λ—the product of identity persistence, cognitive capacity, and social feedback channel fidelity—exceeds one. For a repertoire of such schemas, roles emerge with increasing gain in a bifurcation cascade whose functional form is fixed by the repertoire's eigenvalue spectrum, ranging from monitorable logarithmic sequences to avalanches that arrive without warning. Resource accumulation turns schema competition into a replicator dynamics that, in mean field, selects the cascade type endogenously. Subcritical behavioral covariance reveals that type before onset, which enables early detection, and feedback channel parameters bias which type is selected. Platform design then becomes a control lever to throttle emergent coordination, for example to mitigate risks from deployed AI populations. Joining game theory, collective dynamics, and information engines, we open a route to an information thermodynamics of agent populations.

I. INTRODUCTION

A century-old question runs through the social sciences: how does macro-level social structure shape individuals while, simultaneously, individuals' choices generate that structure? Giddens proposed *structuration* as a resolution to this *structure–agency debate* based on the *duality of structure* [1]: social structure is both the medium and the outcome of the schemas and resources it organizes. Soon after, another sociologist, Sewell, noted that Giddens's duality accounted only for the reproduction of schemas—essentially a local stability condition rather than an endogenous dynamics. Motivated to provide a qualitative account of schema transformation, Sewell recast the duality as a feedback process between *schemas*, the transposable rules and conventions that give meaning to resources by prescribing behavior around their use, and *resources*, the material and symbolic stocks that schemas mobilize and that empower the agents using them [2]. Each sustains the other: schemas direct the use of resources, and resource-wielding agents reproduce (or transform) schemas. Debates resonant with the same micro–macro tension, many unresolved, recur across social science (*e.g.* in cultural evolution [3] and in macroeconomics [4, 5]). An explicit dynamics of structuration would therefore change how those debates are framed.

Despite decades of agent modeling, Sewell's account of resource-schema structuration has no quantitative formalization. Existing agent modeling frameworks do not operationalize its core objects. Opinion dynamics carries a state variable per agent and no resource that accumulates. Cultural evolution uses transmission and replicator models (*e.g.* evolutionary game theory, including spatial reaction-diffusion extensions) that take the trait space or the game as given, without resource accumulation. Agent-based computational economics does accumulate resources under institutions [6], but its institutions are rules imposed on the agents rather than objects the dynamics selects. More specialized models treat resource accumulation [7] and payoff-dependent network reshaping [8], but represent schemas as scalar state or payoff terms rather than as semantically loaded, transposable rules. None of these treats the resource-schema dynamics central to Sewell's vision.

Large language models have opened a new class of applications for agent modeling, both of human social systems [9] and of deployed AI agent systems [10]. Calls for applications of collective dynamics and systems theory from physics [11, 12] now have responses: impressive applications of established frameworks, among them the statistical mechanics of opinion dynamics [13] and replicator-like mean field theories [14]. Structuration, however, has no such framework to carry across. Supplying one now would support a variety of timely research questions on deployed AI, including agent personas and coordination.

We provide such a theory, exemplified for role differentiation in anti-coordination games, and we find that its natural language is thermodynamic. First demonstrated by Szilard almost 100 years ago [15], informa-

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tion engines—systems that convert measurement into extractable work through feedback [16, 17], subject to performance limits set by measurement quality [18]—quantify the value of information for physical agents. We show that a population of agents incentivized and provided with the means to role-play via identity signals constitutes a *collective, self-fueling information engine*: coordinated role-play is the payoff-generating output—the engine's analog of extracted work—that is bounded by the measured role information, just as Szilard's bound limits the work extracted by his illustrative model engine. As an instance of the long-noted link between Maxwell's demon and symmetry-breaking order parameters [19], this description belongs to informational active matter research: a modern expansion of Szilard's ideas to more complex settings, where measurement-informed decisions produce collective order bounded by the information acquired [20, 21]. Those engines reach *consensus*, a magnetization-like order. The engine here produces *differentiation*, a covariance-like order in which agents take complementary rather than matching roles. Social role structure is the engine's running state, sustained by the burning of role information generated by that structure.

We formalize the feedback itself: for a given repertoire of schemas, which become institutionalized, in what order, and which agents are enriched are endogenous outcomes, while what a schema is *about* is held fixed. Transformation in the strict sense requires in addition a mechanism holding schemas on distinct axes, which the game does not fix. The available choices differ in their assumptions and yield a family of theories (Appendix D). In this paper, we center the structuration that every such theory contains by scoping down onto a dynamics of the schema strengths for a fixed repertoire, in which the schema–resource loop already closes.

Our main results are: (i) a minimal model that couples schemas and resources through agent identities and actions, (ii) an ignition condition—$\Lambda > 1$, where dimensionless loop gain Λ factorizes into identity persistence, agent cognitive capacity, and social feedback channel fidelity—for self-sustaining role differentiation; (iii) an extension to multiple schema contexts results in a bifurcation cascade in which role dimensions activate sequentially, with functional form determined entirely by the eigenvalue spectrum of the context repertoire; (iv) an endogenous theory of that spectrum: resources convert schema competition into a replicator dynamics that, in the near-threshold mean field, selects among cascade classes, from single avalanches to logarithmic cascades; and (v) a monitoring protocol—subcritical spectral inversion of the behavioral covariance—that reads out the future cascade of the current repertoire before its onset, subject to finite-sample resolution and slow schema drift. Two ingredients suffice for roles to emerge: identities that persist and are legible to others—lossy summaries of context-dependent action histories—and schema selection driven by the resource-weighted covariance of those identities. The roles they produce are a collectivist solution to the coordination dilemma [22], providing more payoff per agent than the individually rational equilibrium (Sec. II). We close by applying the monitoring framework to mitigate the risks associated with distributional AGI [23]: the hypothesis that general-level capability may emerge from a coordination phase transition among many individually limited AI agents.

II. A MINIMAL ENGINE OF SOCIAL STRUCTURE

The game. We design utility into roles minimally, using a generic anti-coordination game. Two players choose Go or Defer: both playing Go is costly to both $(-1, -1)$, both deferring is mediocre $(3, 3)$, and complementary role-play pays best on average, though asymmetrically $(6, 2)$. The latter is played in a correlated equilibrium that pays 4 per player, while the mixed Nash equilibrium pays only 2.5. We realize this correlated equilibrium using a role signal as the correlating device. Even though role-following is not a dominant strategy—it is obeyed only when partners also follow—its incentive constraint is strict once roles differentiate and agents seek to maximize their payoffs ([24, Sec. S1 C]). The maximal coordination gain $w_0 = 1.5$ per encounter is the work available to an engine that assigns complementary roles through the measurement of social signals. The specific numbers are inessential: any anti-coordination game enters the theory only through this single scalar w_0, which in turn sets merely the scale of wealth. What is essential is the anti-coordination *class*, in which roles are incentive-compatible: each action must be the best reply to its opposite—given Defer, Go must pay more $(6 > 3)$, and given Go, Defer must pay more $(2 > -1)$. Even equal off-diagonal payoffs would suffice. We set them unequal to keep the general case.

Agents, game contexts, and identities. N_a agents play the game against each other in pairwise encounters across a repertoire of M distinct game contexts. We emphasize the importance of variable context in capturing both the transposability of schemas and the context-dependent nature of roles. We represent identities, $\boldsymbol{\omega}_i$ for agent i, and contexts, $\mathbf{g}_m$ for context m, as vectors $\boldsymbol{\omega}_i, \mathbf{g}_m \in \mathbb{R}^d$ with embedding dimension $d \gg 1$. Each agent stores and presents its own identity in each encounter, though a public registry of identities is an equivalent alternative implementation (Sec. III A). For simplicity, the identity of agent i in context m is contextualized as the scalar $\boldsymbol{\omega}_i \cdot \mathbf{g}_m$. When paired with j in an encounter, agent i makes a noisy observation of the role-signal $s = \Delta_{im}(j) + \sigma_\mathrm{obs}\xi$, with signed difference $\Delta_{im}(j) = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j) \cdot \mathbf{g}_m$—its sign prescribes the role, its magnitude sets how reliably that role can be inferred—and ξ a standard Gaussian noise source. This reading is the population's single *status channel*. The context fixes only which projection of identity it carries. Averaged over the drawn partner j, this reduces to Δ_{im} below. The policy of the correlated

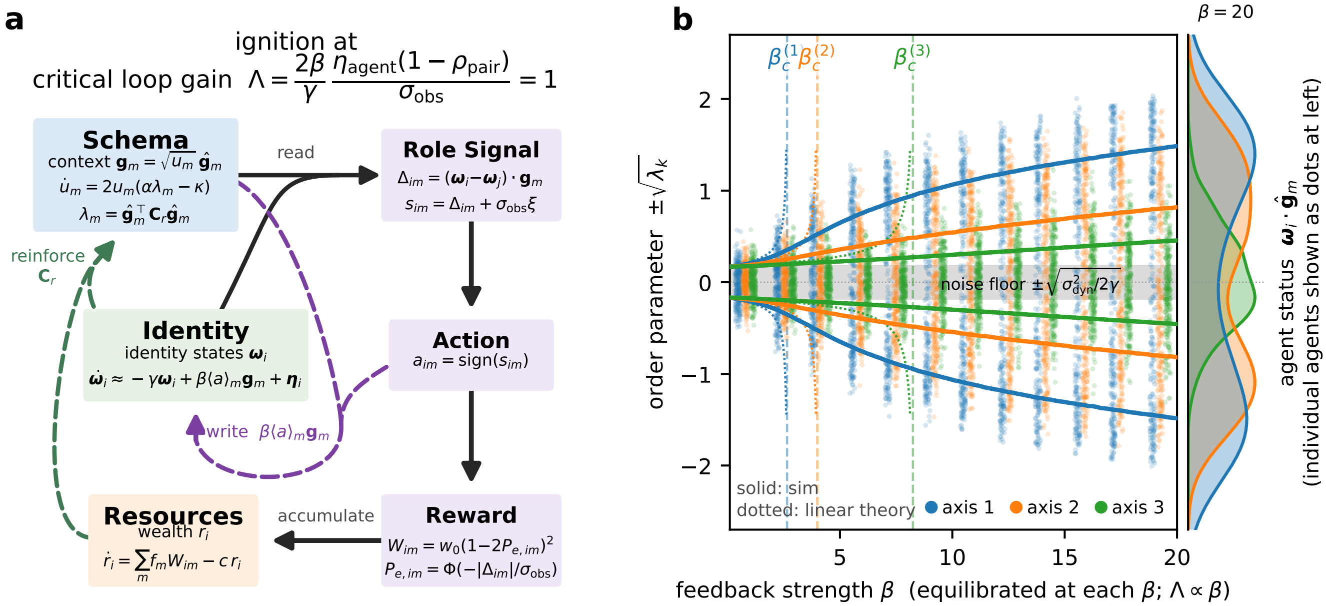

FIG. 1. **The self-consistent schema–role–resource loop and its ignition.** **(a)** Model schematic, separating the three persistent state variables (schemas $\mathbf{g}_m = \sqrt{u_m}\,\hat{\mathbf{g}}_m$, of which only u_m evolves; identities $\boldsymbol{\omega}_i$; resources r_i) from the fast per-encounter chain (status, action, reward). *Solid* arrows are the feed-forward read-out: schema and identity form the status signal, the policy thresholds it into an action, and the coordination reward accumulates as resources. *Dashed* arrows are the slow state updates that close the loop: resources reweight identities to reinforce schemas through the resource-weighted covariance $\mathbf{C}_r$, and actions are inscribed back into identities along the same axis (the write $\beta\langle a\rangle_m \mathbf{g}_m$). The single direction $\mathbf{g}_m$ is thus both read and write axis, which makes Sewell's schema–role duality a fixed-point loop, and the engine ignites when the loop gain $\Lambda > 1$ (top). **(b)** Three sequential bifurcations (order parameter vs. coupling) over the β sweep ($N_{\mathrm{a}} = 2500$, $d = M = 3$, hierarchical fixed context repertoire, random pairing; Appendix B). Points are individual agent statuses $\boldsymbol{\omega}_i \cdot \hat{\mathbf{g}}_m$ along the three schema axes. Solid curves are the order parameter $\pm\sqrt{\lambda_k}$, the two branches of a pitchfork on each axis, with λ_k the k-th eigenvalue of the population covariance $\mathbf{C}$. Below its critical coupling an axis is unimodal within the noise band $\pm\sqrt{\sigma_{\mathrm{dyn}}^2/2\gamma}$; above it (dashed) it splits into the complementary leader/follower role modes of the anti-coordination game, which separate further with β. Dotted curves are the linear mean-field theory, diverging at $\beta_c^{(k)}$ (Eq. (5)) while the quartic confinement saturates the simulated branch—the visible cascade signature. Axis 1 splits first, then 2, then 3. The right strip shows the status distribution at $\beta = 20$, axis 1 fully bimodal, axis 3 only marginally. Left and right axes are the same role projection, read as order parameter and as agent status. Schemas are held fixed to display the cascade cleanly. The full loop selects this ordering endogenously (Fig. 3).

equilibrium prescribes Go ($a = +1$; payoff 6) when s is positive and Defer ($a = -1$; payoff 2) when negative. To reflect the unequal payoffs, we call s relative *status* (as in 'social status'), though we emphasize that status is relative to the game context. Integrating T noisy samples, the Bayes-optimal decision is deterministic: $a = \mathrm{sign}(\frac{1}{T}\sum_t s_t)$.

Roles require persistent identities. We take a simple choice: each identity low-pass filters its agent's action history, here a context-dependent signed action input. This evolves under a restoring force and write noise,

$$\dot{\boldsymbol{\omega}}_i = -\gamma\,\boldsymbol{\omega}_i(1+|\boldsymbol{\omega}_i|^2) + b_0 \sum_{e\in E_i} a_e\,\mathbf{g}_{m_e}\,\delta(t-t_e) + \sigma_{\mathrm{dyn}}\boldsymbol{\eta}_i, \tag{1}$$

where E_i is the set of encounters involving agent i, a_e and m_e the decision and context of encounter e, b_0 the per-encounter kick, γ the decay rate of identity, and $\boldsymbol{\eta}_i$ a vector of standard Gaussian noise sources. This intrinsic write noise captures the effects of unobserved degrees of freedom beyond the encounters. Equation (1) is an overdamped Langevin equation in a quartic confining potential: the restoring force is $-\nabla_{\boldsymbol{\omega}_i}\gamma(|\boldsymbol{\omega}_i|^2/2 + |\boldsymbol{\omega}_i|^4/4)$, whose cubic term serves only to keep $|\boldsymbol{\omega}_i|$ finite (Appendix A). Its coefficient is tied to γ as a choice of units for $|\boldsymbol{\omega}_i|$, absorbed by rescaling $\mathbf{g}_m$ and β, so the linear decay rate is the only free parameter of the restoring force. Encounters are Poisson at rate ν, drawn across contexts with normalized frequencies $\sum_m f_m = 1$ and over partners with pairing probabilities $p_{ij}^{(m)}$. Averaging each decision over observation noise (giving the expected signed action $\langle a\rangle_m$), over the drawn context and partner, taking the mean field ($N_{\mathrm{a}} \to \infty$), and coarse-graining the Poisson kick stream in its dense-encounter diffusion limit

($\nu \to \infty$, $b_0 \to 0$ at fixed $\beta \equiv b_0\nu$), the drift becomes

$$\dot{\boldsymbol{\omega}}_i = -\gamma\,\boldsymbol{\omega}_i(1+|\boldsymbol{\omega}_i|^2) + \beta \sum_m f_m \,\langle a\rangle_m \,\mathbf{g}_m + \sigma_{\rm dyn}\boldsymbol{\eta}_i, \quad (2)$$

where $\langle a\rangle_m$ is the expected signed action in context m—a sigmoid of the deterministic status difference, Δ_{im}. Writing $\pi(\Delta_{im}) = \Pr(a{=}{+}1|\langle s\rangle_\xi = \Delta_{im})$ for the noise-driven Go-probability, we have $\langle a\rangle_m = 2\cdot\pi(\Delta_{im}) - 1$, so its slope at zero difference is $2\pi'(0)$—the read gain that sets how strongly actions are written into memory (Bayesian policy on Gaussian noise gives $\pi'(0) = \sqrt{T/2\pi}/\sigma_{\rm obs}$). For simplicity, we consider well-mixed pairing, which averages the partner projection to the population mean. This vanishes in the symmetric state, the undifferentiated population before any role structure exists, so the difference and thus $\langle a\rangle_m$ depend on $\boldsymbol{\omega}_i$ alone. For structured pairings, the effect enters through $p_{ij}^{(m)}$ as a graph Laplacian (see Appendix A for extensions).

Resources and schemas. Coordination income accumulates as agent wealth,

$$\dot{r}_i = \sum_m f_m W_{im} - c\,r_i, \ \text{ with } \ W_{im} = w_0(1-2P_{e,im})^2 \quad (3)$$

the payoff above Nash at role-inference error rate $P_{e,im}$ ([24, Eq. (S2)]) and c the resource-decay rate, so $1/c$ is the horizon over which past income still counts. This payoff obeys a Szilard bound, $W \le 2\ln 2\, w_0\, I$ with $I = 1 - H_{\rm bin}(P_e)$ the role information a measurement supplies ([24, Sec. S1]). This is the differentiation case counterpart of the information bound on consensus order case in active matter [20]. The bound is on the coordination payoff, in information units rather than as a thermodynamic budget (Appendix A). The symmetric form $I = 1 - H_{\rm bin}(P_e)$ is the equiprobable two-role idealization. Heterogeneous status differences and pairwise error rates enter through the partner average W_{im} that follows. Here W_{im} is a partner-average of the pairwise payoff $W_{ijm} = w_0(1-2P_{e,ijm})^2$, set by the magnitude of the status difference to the partner and averaged over the same pairing statistics $p_{ij}^{(m)}$ that enter Eq. (1) (extensions treated in [24, Sec. S2]).

Agents who profit from taking differentiated roles along a schema reinforce it. A schema is a vector, on whose two parts reinforcement acts separately: its direction $\hat{\mathbf{g}}_m$ fixes what the schema is *about*, its magnitude how strongly that prescription is *institutionalized*. Writing $\mathbf{g}_m = \sqrt{u_m}\,\hat{\mathbf{g}}_m$ with $u_m = |\mathbf{g}_m|^2$ splits any such dynamics exactly into a *strength sector* carrying u_m—the reinforcement of a schema already in the repertoire—and a *transformation sector* rotating $\hat{\mathbf{g}}_m$, the identity space direction along which anti-correlation in the paired agents' actions confers coordination benefit.

We develop the strength sector, which suffices for schema-resource duality: it determines which of an available repertoire become institutionalized, in what order, and which agents are enriched. The $\{\hat{\mathbf{g}}_m\}$ are then fixed as environmental: the repertoire set of games available to be played. The environment also fixes where the schemas start: $u_m(0) = u_0 e^{\sigma_u z_m}$, with u_0 the seed scale, σ_u the log-dispersion across contexts, and z_m standard normal. The dispersion σ_u fixes how far apart in time successive contexts cross threshold (Sec. V C), and for a deployed system it is set by the platform (Appendix C). Restricting a dynamics to magnitudes along a fixed frame is standard practice—amplitude equations for a fixed set of critical modes [25], replicator and Lotka–Volterra dynamics on a fixed set of types [26], the overlap coordinates of associative memories with quenched patterns [27], and the decoupled mode dynamics of deep linear networks under whitened inputs [28] all take this form. Here the restriction is exact rather than approximate: Eq. (4) is one half of a polar split of the vector dynamics. The other directional half, dynamics of the transformation sector, is analyzed in Appendix D where we show repertoire diversity depends strongly on what additional assumptions are made about how schemas interact.

A schema's strength u_m acts at each stage of an encounter with it. It scales the status difference read, $\Delta_{im} \propto \sqrt{u_m}$, which lowers the role-inference error rate $P_{e,im}$ against fixed noise $\sigma_{\rm obs}$. It also fixes the social gain $2\pi'(0)u_m$ and hence the coupling at which context m bifurcates (Sec. III A). Contexts therefore activate one at a time as the coupling rises, in the order of their strengths u_m. This sequence of bifurcations, the *cascade* (Sec. IV), is the focus of much of the rest of the paper. The strengths also draw on one finite collective capacity to signal and read status, $\sum_m u_m \le U_{\rm cap}$ (Sec. V B), so once the total reaches it, raising one lowers the rest. Institutionalization is thus the allocation of that capacity across the available games. The dynamics below is the rule by which coordination success reallocates it.

Each strength then grows in proportion to how much resource-weighted structure lies along its own axis, and decays at a rate common to every schema:

$$\dot{u}_m = 2u_m\big(\alpha\lambda_m - \kappa\big), \qquad \lambda_m = \hat{\mathbf{g}}_m^\top \mathbf{C}_r\,\hat{\mathbf{g}}_m,$$
$$\mathbf{C}_r = \frac{\sum_i r_i\,\boldsymbol{\omega}_i\boldsymbol{\omega}_i^\top}{\sum_i r_i} \quad (4)$$
$$\text{subject to } \ U \le U_{\rm cap}\,, u_m \le u_{\rm cap}$$

where $\mathbf{C}_r$ is the resource-weighted population covariance, λ_m the resource-weighted variance along context m, κ the context decay rate, and α the context reinforcement rate. We impose two capacity constraints, both properties of that channel. All M schemas are read through the one status channel, so they draw on one budget rather than holding one each. Because the status difference read on axis m scales as $\sqrt{u_m}$, the strength u_m is the power that axis puts on the channel, and the total $U = \sum_m u_m$ is the power the population spends signaling status at once. A channel of finite power caps that total at $U_{\rm cap}$. A channel of finite range caps any single axis at $u_{\rm cap}$, since status can be displayed only up to the largest difference the channel represents. Power and range are separate properties of one channel, so their ratio is free, and it decides which limit is reached first (Sec. V C). Equivalently $d\ln u_m/dt = 2(\alpha\lambda_m - \kappa)$, the factor of two because $u_m = |\mathbf{g}_m|^2$ grows at twice the rate of $\mathbf{g}_m$. The two ingredients of λ_m encode Sewell's duality. It is *covariance*-driven because a schema can organize only the behavioral variation that exists: only the differentiation already present along axis m can be divided into complementary roles, so a schema is reinforced along the axes of differentiation the roles themselves produced. A schema aligned with no variance prescribes nothing and decays at rate κ. It is *resource*-weighted because a schema is validated by the payoff its adherents accrue rather than by raw differentiation: weighting by wealth r_i makes reinforcement contingent on success—the operational form of Sewell's claim that resources, accumulated by enacting a schema, sustain the schema that produced them. With reinforcement proportional to u_m, Eq. (4) selects among existing schemas rather than creating new ones, at a per-unit rate set by λ_m. Because the decay is common to all schemas, the shares u_m/U obey a replicator equation with fitness $\alpha\lambda_m$ exactly. The finite capacity regulates the total (Sec. V B). Transposability enters as the coherence of the repertoire, the overlaps $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$, through which differentiation along one axis raises the resource-weighted variance along another (Sec. V B).

Equations (1), (3) and (4) together form the minimal social technology needed to close the perception-action-reward loop (Fig. 1a) in a way sufficient for social roles to emerge. The resulting multi-agent system dynamics formalizes Sewell's vision and generates duality as a self-consistent fixed-point equation: schemas shape the role structure, roles generate resources, and resource-weighted structure ($\mathbf{C}_r$) selects schemas. The resource weighting makes survival contingent on success: a schema whose axis carries differentiation among agents who profit by it grows, while one whose adherents gain nothing decays at rate κ however much behavior varies along that axis (Sec. V A). The solution set of these equations of motion depends on the parameters and does not generically produce roles. Thus, in the next section we determine for a given game repertoire the nature and location of the phase transition across which role-based society emerges.

III. PARAMETRIZED IGNITION

By linearizing Eq. (2) about the symmetric state, in this section we show that the expected social feedback to the identity is $\beta(1-\rho_{\text{pair}})\,\mathbf{G}\,\boldsymbol{\omega}_i$ with gain matrix $\mathbf{G} = 2\pi'(0)\,\boldsymbol{P}^\top\boldsymbol{P}$, where $\boldsymbol{P}$ stacks the schema vectors $\mathbf{g}_m$ and $(1-\rho_{\text{pair}})$ is the pairing gain factor—set by the matching scheme's partner correlation ρ_{pair} (defined below), equal to 1 for random matching and up to 2 for status-mirrored. The symmetric state destabilizes along eigenvector k of $\mathbf{G}$ when β exceeds

$$\beta_c^{(k)} = \frac{\gamma}{(1-\rho_{\text{pair}})\mu_k}, \tag{5}$$

where $\mu_1 \geq \cdots \geq \mu_R > 0$ are the non-zero eigenvalues of $\mathbf{G}$ ($R = \operatorname{rank}(\boldsymbol{P})$): role axes activate one at a time as the feedback strength grows. We make the population covariance $\mathbf{C} = N_{\text{a}}^{-1}\sum_i \boldsymbol{\omega}_i\boldsymbol{\omega}_i^\top$ the order parameter. Its eigenvalues λ_k leave the noise floor $\sigma_{\text{dyn}}^2/2\gamma$ one at a time. The number that have crossed, the height of the cascade $N(\beta)$, is the *cultural thickness*: the count of active (supra noise floor) role dimensions.

For a single axis the ignition condition $\beta G > \gamma$ is a loop-gain criterion which factorizes as

$$\Lambda \equiv \underbrace{\frac{2\beta}{\gamma}}_{\substack{\text{identity}\\ \text{persistence}}} \cdot \underbrace{\eta_{\text{agent}}}_{\substack{\text{cognitive}\\ \text{capacity}}} \cdot \underbrace{\frac{1-\rho_{\text{pair}}}{\sigma_{\text{obs}}}}_{\substack{\text{channel}\\ \text{fidelity}}} > 1, \tag{6}$$

where $\eta_{\text{agent}} \equiv \sigma_{\text{obs}}\,\pi'(0)$ is the agent's dimensionless inference efficiency ($1/\sqrt{2\pi}$ for a threshold agent, $\sqrt{T/2\pi}$ with memory T) and $\rho_{\text{pair}} \in [-1,1]$ is the correlation the matching scheme induces between a partner's status and one's own (at fixed context m). The pairing enters the loop gain only through the factor $(1-\rho_{\text{pair}})$. Independent (random) pairing gives $\rho_{\text{pair}} = 0$, factor 1—the reference, recovering the mean-field threshold $\beta_c = \gamma/\mu$; rank-ordered (status-mirrored) matching drives $\rho_{\text{pair}} \to -1$ in the population limit, factor 2, halving the threshold; assortative (homophilic) matching gives $\rho_{\text{pair}} \to 1$, factor 0 ([24, Sec. S2]). The correlation needs no pre-existing role modes. ρ_{pair} is a property of the matching algorithm applied to whatever status distribution currently exists, including a unimodal and arbitrarily narrow one, so it is as well defined below threshold, where it sets β_c, as above it.

The engine runs on two integrators—a write path (β/γ: decisions accumulated into persistent identity) and a read path (η_{agent}: social signals accumulated into posterior inference)—coupled by the information channel between agents. The factors compensate at linear order: well-designed matching can compensate for limited-inference agents and vice versa (similar to the measurement-quality bound on engine performance derived in [18]). However, even at the best (rank-ordered) matching ($\rho_{\text{pair}} \to -1$) ignition requires $\eta_{\text{agent}} > \sigma_{\text{obs}}\gamma/4\beta$, an *irreducible cognitive floor* below which no interaction orchestration can sustain role structure. At threshold where linear order is accurate, only the product Λ is physical. Beyond this threshold the degeneracy of the factorization breaks: β sets how far the roles separate, independently of ρ_{pair}, while ρ_{pair} sets how often that separation converts to payoff. Role societies at equal Λ but different splits are therefore dynamically distinguishable ([24, Sec. S2]). Figure 1b shows the first three bifurcations in the cascade: equilibrating the population at each β, role axes leave the noise floor one at a time around their critical coupling strengths $\beta_c^{(k)} = \gamma/[(1-\rho_{\text{pair}})\mu_k]$.

A. The gain matrix and the cascade of critical couplings

We now linearize the mean-field drift Eq. (2) about the symmetric state, showing how the gain matrix $\mathbf{G} = 2\pi'(0)\boldsymbol{P}^\top\boldsymbol{P}$ arises. Its feedback term is $\beta\sum_m f_m\langle a\rangle_m\mathbf{g}_m$, with $\langle a\rangle_m(\boldsymbol{\omega}_i)$ the expected signed action in context m, averaged over the observation noise and the drawn partner (Sec. II).

Mean field. As $N_\mathrm{a}\to\infty$ the partner projection $\boldsymbol{\omega}_j\cdot\mathbf{g}_m$ is replaced by its population mean, which vanishes at the symmetric fixed point, so $\Delta_{im}(j)\to\Delta_{im}\equiv\boldsymbol{\omega}_i\cdot\mathbf{g}_m$, $s_{im}\approx\Delta_{im}+\sigma_\mathrm{obs}\xi_{im}$, and, averaging the decision over the observation noise,

$$\langle a\rangle_m(\boldsymbol{\omega}_i) = 2\pi(\Delta_{im}) - 1, \tag{7}$$

which for the naive policy is $\operatorname{erf}\big(\Delta_{im}/(\sigma_\mathrm{obs}\sqrt{2})\big)$.

Linearization. At $\boldsymbol{\omega}_i=\mathbf{0}$ each projection vanishes and $2\pi(0)-1=0$ by policy symmetry. To first order $\langle a\rangle_m\approx 2\pi'(0)(\mathbf{g}_m^\top\boldsymbol{\omega}_i)$, so the feedback term of Eq. (2) becomes

$$\beta\sum_m f_m\langle a\rangle_m\mathbf{g}_m \approx 2\beta\pi'(0)\Big(\sum_m f_m\,\mathbf{g}_m\mathbf{g}_m^\top\Big)\boldsymbol{\omega}_i = \beta\,\mathbf{G}\,\boldsymbol{\omega}_i, \tag{8}$$

$$\mathbf{G}\equiv 2\pi'(0)\sum_m f_m\,\mathbf{g}_m\mathbf{g}_m^\top = 2\pi'(0)\,\boldsymbol{P}^\top\operatorname{diag}(\mathbf{f})\,\boldsymbol{P}, \tag{9}$$

with $\mathbf{f}=(f_1,\ldots,f_M)$ and $\boldsymbol{P}$ the $M\times d$ matrix of stacked context vectors. We take the play frequencies uniform, $f_m=1/M$, and absorb the constant $1/M$ into β, so that $\mathbf{G}=2\pi'(0)\,\boldsymbol{P}^\top\boldsymbol{P}$ and β is the per-context write rate. The frequencies are retained explicitly only where they act as a design lever ([24, Sec. S2 D]). They are properties of the environment, fixed over the full set of M candidate contexts, and are not renormalized as schemas go extinct. Combined with the linearized restoring force (the cubic term vanishes at the origin),

$$\dot{\boldsymbol{\omega}}_i \approx (-\gamma I+\beta\mathbf{G})\boldsymbol{\omega}_i + \sigma_\mathrm{dyn}\boldsymbol{\eta}_i. \tag{10}$$

The symmetric state loses stability along eigenvector k of $\mathbf{G}$ when β exceeds $\gamma/[(1-\rho_\mathrm{pair})\mu_k]$ (Eq. (5)). Absorbing the constant $(1-\rho_\mathrm{pair})$ into β as above,

$$\beta_c^{(k)} = \gamma/\mu_k, \qquad \mu_1\ge\mu_2\ge\cdots\ge\mu_d, \tag{11}$$

the cascade of critical couplings.

Validity of the linearization. The subcritical state is not the point $\boldsymbol{\omega}=0$ but a Gaussian cloud of finite width about it (covariance $\mathbf{C}$). Stability, however, is a Jacobian property evaluated *at* the fixed point, independent of where the probability mass sits, so Eq. (11) is the exact stability boundary at any noise level. The cloud's width enters only through the partner average, which softens the gain and shifts onset by $O(\sigma_\mathrm{dyn}^2)$. This is a mean-field Ginzburg correction that vanishes with σ_dyn and is negligible outside a shrinking critical window ([24, Sec. S6]).

IV. THE FUNCTIONAL FORM OF THE CASCADE

We assume role-based societies emerge from a ramping of feedback strength with time, $\beta(t)$, summarizing the culture and technology innovations that improve the fidelity of inter-agent communication. We want to understand the resulting bifurcation cascade in which differentiated roles emerge within an increasing number of game contexts. In this section, we consider the case where the ramp satisfies separation of timescales, $\kappa\ll\dot{\beta}/\beta\ll\gamma$. This is fast compared with the schema dynamics, so the spectrum of the social gain matrix is quasi-static across a sweep and we can classify cascades over β (not time; $\beta(t)$ is treated only in the monitoring analysis (Sec. VI)). It is slow compared with agent relaxation, so the population tracks it. Where these two separations hold, the spectrum is the same at every point of the sweep, so the cascade is the readout of one game repertoire (directions and strengths). Where they fail, mode k approaches its threshold at the combined rate $\dot{\beta}/\beta + d\ln u_k/dt$: a drift common to every mode rescales that rate uniformly and leaves the shape of the cascade intact (so the treatment in this section is still exhaustive). Alternatively, a spread across modes moves the thresholds relative to one another during the sweep, and the observed cascade need not match any single class. We do not treat that case (Sec. VI A) in this paper.

The number of active axes at coupling β is the counting function $N(\beta)$. Here, we derive the effective local rate of growth of this function as that for the eigenvalue spectrum ρ of the social gain matrix $\mathbf{G}$ in the limit $M,d\to\infty$ at fixed aspect ratio M/d, $-d\ln\rho/d\ln\mu$. We find that only three repertoire properties shape $N(\beta)$: the strength hierarchy $\{u_m\}$, the coherence $\{\hat{\mathbf{g}}_m\cdot\hat{\mathbf{g}}_n\}$, and the aspect ratio M/d. Equation (4) makes the first endogenous and leaves the other two to the environment. These generate five classes (Fig. 2): degenerate (single avalanche, no warning), geometric (constant rate), Zipf with exponent $a\gtrless 1$ (decelerating versus runaway acceleration), random Marčenko–Pastur bulk (square-root-slow onset, then burst) [29], and spiked (early bifurcation, quiescent interval, avalanche). Figure 2a gives an example of each for $M=50$, while we show the rate for $M\to\infty$ in Fig. 2b.

A. The counting function and acceleration criterion

Let $\mu_1\ge\cdots\ge\mu_R>0$ be the nonzero eigenvalues of $\mathbf{G}$, $R=\operatorname{rank}(\boldsymbol{P})\le\min(M,d)$. Axis k bifurcates at $\beta_c^{(k)}=\gamma/[(1-\rho_\mathrm{pair})\mu_k]$ (Eq. (5)). Absorbing the fixed factor $(1-\rho_\mathrm{pair})$ into β for this section, as with the frequencies above, this is $\beta_c^{(k)}=\gamma/\mu_k$. The number of active axes at coupling β is

$$N(\beta) = \#\{k:\mu_k>\gamma/\beta\} = R\,\bar{F}(\gamma/\beta), \tag{12}$$

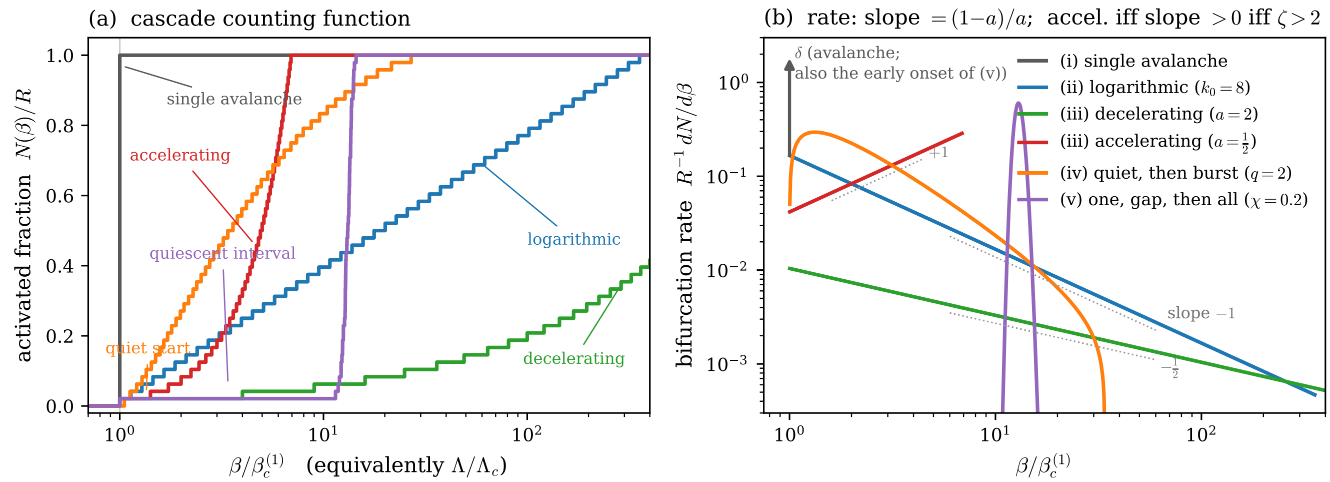

FIG. 2. **Cascade classes of exogenous context repertoires.** **(a)** The number of active role axes $N(\beta)$. **(b)** The bifurcation rate. Both are shown over the five spectral classes of the context Gram matrix (legend): (i) degenerate spectra give a single avalanche with no advance warning (gray); (ii) geometric hierarchies give a logarithmic cascade with constant bifurcations in log-space (blue); (iii) power-law (Zipf) hierarchies split at exponent $a = 1$ into decelerating (informative early phase; green) and accelerating (early rate under-predicts; red) cascades; (iv) unstructured random repertoires (Marčenko–Pastur; orange) start quietly, $dN/d\beta \propto (\beta - \beta_c)^{1/2}$, then burst; (v) coherent (spiked) repertoires (purple) produce an early bifurcation, a quiescent interval, then an avalanche.

$$\bar{F}(\mu) = \int_{\mu}^{\infty} \rho(\mu')\, d\mu', \tag{13}$$

with ρ the spectral density of $\mathbf{G}$: the cascade is the push-forward of the spectrum under $\mu \mapsto \gamma/\beta$. The rate is

$$\frac{dN}{d\beta} = R\frac{\gamma}{\beta^2}\rho\Big(\frac{\gamma}{\beta}\Big), \qquad \frac{dN}{d\ln\beta} = R\,\mu\rho(\mu)\big|_{\mu=\gamma/\beta}. \tag{14}$$

Defining the local spectral exponent $\zeta(\mu) \equiv -d\ln\rho/d\ln\mu$, differentiating again gives the acceleration criterion

$$\frac{d^2N}{d\beta^2} > 0 \iff \frac{d}{d\mu}[\mu^2\rho(\mu)] < 0 \iff \zeta > 2 \text{ at } \mu = \gamma/\beta. \tag{15}$$

A monitor extrapolating a low early rate implicitly assumes $\zeta \leq 2$ in the unreached part of the spectrum. Whether that holds is a computable property of the repertoire. Only three repertoire properties shape ρ: the strength hierarchy $\{u_m\}$, the coherence $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, and the aspect ratio $q = M/d$.

B. The five exogenous classes

Table I catalogs the qualitatively distinct spectral classes and the cascade each induces. Any repertoire's spectrum is locally one of these, and a general cascade concatenates them. Each class is the canonical form taken when one of the three properties dominates. The strength hierarchy $\{u_m\}$ alone sets μ_k when contexts are effectively orthogonal, and its functional form—flat, exponential, or power-law—distinguishes classes (i)–(iii). The aspect ratio $q = M/d$ alone shapes class (iv), whose Marčenko–Pastur edges are functions of q. The coherence $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, through its uniform value χ, produces class (v)'s spike-plus-bulk split, which a strongly condensed strength hierarchy also produces at zero coherence.

The boundary $a = 1$ ($\zeta = 2$) separates the decelerating and accelerating Zipf regimes. The random class [29] has its spectrum confined to $[\mu_-, \mu_+]$, with the edges set by the aspect ratio $q = M/d$. Bifurcations run from the largest eigenvalue down, so the cascade enters the bulk at its upper edge, where the density vanishes as $\rho \sim \sqrt{\mu_+ - \mu}$—hence the vanishing onset rate $dN/d\beta \propto (\beta - \beta_c)^{1/2}$ before the bulk. The spiked class, from redundancy (coherent contexts sharing directions), gives one outlier $\mu_{\text{sp}} = |g|^2(1+(M-1)\chi)$ over a degenerate bulk $\mu_\perp = |g|^2(1-\chi)$ at uniform coherence χ, hence a quiescent interval of relative width $(1+(M-1)\chi)/(1-\chi)$ and then an avalanche. Rate data alone cannot distinguish this interval from a cascade that has ended.

V. SCHEMA-RESOURCE DYNAMICS SELECT THE CASCADE CLASS

In the previous section, the schema strengths, and thus the cascade type (Fig. 2), were taken as given. In the model they emerge endogenously from the stationary solution of the schema-resource dynamics. The directions stay exogenous throughout (Sec. II). Here, we analyze that solution space through the lens of the result-

TABLE I. Spectral classes and the cascades they induce. The last column anticipates the monitoring analysis of Sec. VI.

Spectrum	Set by	Cascade	μ_k	$N(\beta)$	For a monitor
(i) degenerate	strengths flat	single avalanche	$\mu_k = \mu$	step	no warning
(ii) geometric	strengths exponential	logarithmic	$\mu_k \propto e^{-k/k_0}$	$k_0 \ln(\beta/\beta_c)$	steady per e-fold
(iii) Zipf $a > 1$	strengths power-law	decelerating	$\mu_k \propto k^{-a}$	$(\beta/\beta_c)^{1/a}$	informative early
(iii) Zipf $a < 1$	strengths power-law	accelerating	$\mu_k \propto k^{-a}$	$(\beta/\beta_c)^{1/a}$	early rate too low
(iv) random	aspect ratio q	quiet, then burst	Marčenko–Pastur	$(\beta - \beta_c)^{3/2}$	loop still open
(v) spiked	strengths spiked, or coherence χ	one, gap, then all	outliers + bulk	gap, then jump	quiet mimics an end

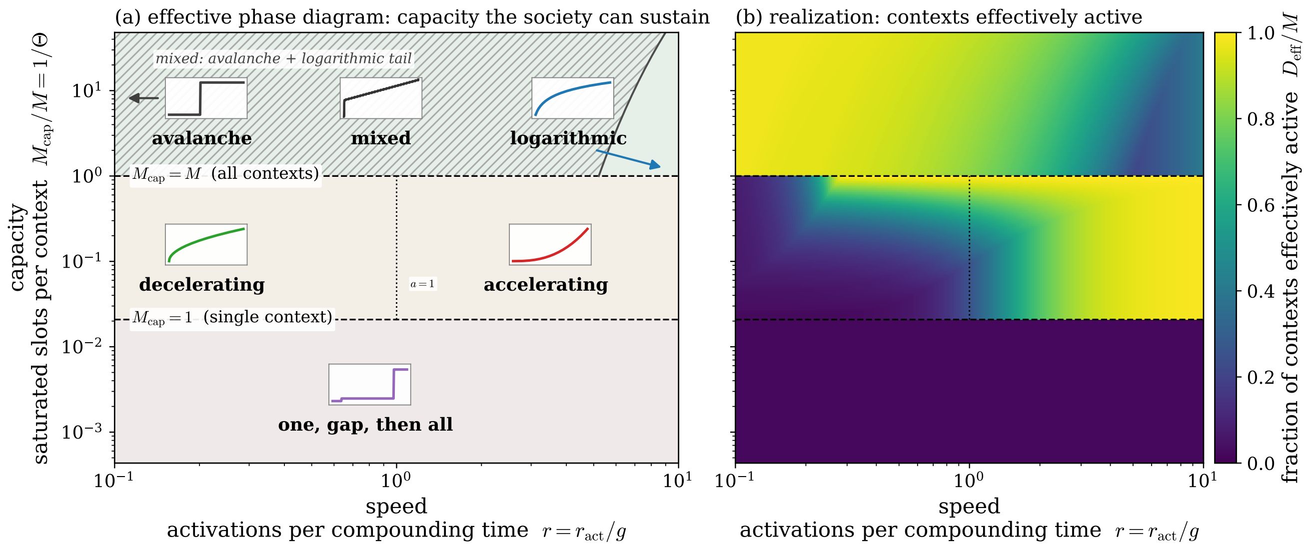

FIG. 3. **Cascade types and thickness over replicator dynamics.** Endogenous cascade regimes over the two coordinates that separate them. The horizontal axis (*speed*) is $r = r_{\mathrm{act}}/g$, the number of axes that activate per compounding time $1/g$, fixed by the dispersion of the seeded repertoire through Eq. (26). The vertical axis (*capacity*) is M_{cap}/M, the saturated slots per context, that is the fraction of the M candidate schemas the finite social capacity can sustain as live contexts ($M_{\mathrm{cap}} = U_{\mathrm{cap}}/u_{\mathrm{cap}}$). **(a)** *Capacity*: the contexts the society *can* sustain, a stationary-state property fixed by $\Theta = M/M_{\mathrm{cap}}$ alone, which splits the plane into three bands with sharp counting-threshold boundaries: all M contexts, a graded cluster of M_{cap}, or one. Inset sketches show the sequence of activations the transient sort passes through, set by r: avalanche to logarithmic in the thick band ($\Theta \leq 1$), decelerating ($a > 1$) to accelerating ($a < 1$) across $a = 1$ in the middle band, and one onset, a gap, then all the rest in the thin band. The left-hand arrow marks $r \to 0$, where the cascade is a pure avalanche. The random Marčenko–Pastur class (iv of Fig. 2) is absent because it is an unorganized initial condition rather than an endpoint the loop selects (Sec. V C). Within the thick band a saturated cluster coexists with a geometric tail (hatched) below the exact boundary $r = M/L(\Theta)$. The right-hand arrow runs from the logarithmic sketch past that boundary to a purely geometric spectrum, and the inset shows $N(\beta)$ at a coexistence point—an early jump as the cluster ignites at once, then the tail's gradual ramp. **(b)** *Realization*: the contexts *effectively active*, the participation ratio $D_{\mathrm{eff}} = (\sum_k \mu_k)^2 / \sum_k \mu_k^2$ of the gain spectrum $\{\mu_k = 2\pi'(0)u_k\}$, normalized by M. It depends on how far the transient sort has progressed, hence on r, and falls below capacity wherever the sort is incomplete—bright on the accelerating side, darker on the decelerating side. The difference between the panels measures distance from the stationary endpoint. Exact boundaries, their mean-field status, the ceiling u_{cap}, and the held-fixed groups ($\alpha\langle\lambda\rangle/\kappa$, $\beta\mu_1/\gamma$, M/d) in Sec. V C.

ing cascade type to arrive at a diagram denoting which regimes of schema-resource dynamics exhibit which cascades types across the transition to a role-based society. In particular, we take the M contexts as an ambient pool of candidate schemas and ask how many survive and with what strengths. The active set is those with $u_m = |\mathbf{g}_m|^2$ above the seed scale at which schemas are initialized, an emergent number $K \leq M$. The threshold is bookkeeping for the transient: at the fixed point the inactive schemas have decayed to zero and K is set by $\alpha\lambda_m \geq \kappa$ alone (Eq. (21)). Rewriting Eq. (4), strengths grow by the resource-weighted variance along each axis, $\frac{d}{dt} \ln u_m = 2(\alpha\lambda_m(\mathbf{C}_r) - \kappa)$. Imposing a finite social capacity $\sum_m u_m \leq U_{\mathrm{cap}}$—the collective attention the population can spend legibly signaling status across all active schemas at once—limits the total. The shares follow a replicator dynamics with per-schema fitness $\alpha\lambda_m$ [30, 31], generated by the engine chain: *schema* $\to$ *differentiation*

$\to$ *signal* $\to$ *coordination* $\to$ *resources.*

The competition has a single interior fixed point, and it is a saddle: stable in total magnitude, where an imposed ceiling regulates the sum, and unstable in *composition*, where a schema that gains strength becomes fitter and grows at the expense of the rest. Section V B derives both directions. The dynamics therefore leave the saddle one axis at a time. An inactive axis sits with its covariance eigenvalue pinned at the noise floor $\sigma^2_{\rm dyn}/2\gamma$. When an axis activates, the resources it generates reweight $\mathbf{C}_r$ toward the agents who differentiated along it. Because those agents also spread in other directions, this inflates the residual variance off the active axis, and the next schema grows until its eigenvalue clears the floor and ignites. Successive axes leave the floor in turn—this is the endogenous cascade.

What are the properties of the replicator dynamics that control which cascade type is exhibited? We consider transient and stationary components in turn. First, the transient, which we call the *sort*: the seeded pool of candidates is winnowed and ordered into a strength hierarchy en route to the stationary state. It has two emergent rates, an *activation rate* $r_{\rm act}$ at which successive axes cross threshold and a *compounding rate* g at which an active schema's strength grows. Both are outputs of the dynamics, not control knobs. Their ratio $r = r_{\rm act}/g$ counts the axes that cross threshold per *compounding time* $1/g$, the time an active schema takes to grow by a factor e. It is set by the dispersion of the seeded repertoire (Sec. V B), that is, by how unequally the candidate contexts are weighted at the outset. The stationary dynamics has four dimensionless quantities, but only one is needed to span the morphology between regimes (the rest set a fixed backdrop). The ratio $\Theta = M/M_{\rm cap}$ is the number of schemas competing per capacity slot, where the *capacity* $M_{\rm cap} = U_{\rm cap}/u_{\rm cap}$ counts how many schemas the finite capacity $U_{\rm cap}$ sustains at the per-schema ceiling $u_{\rm cap}$. Large Θ divides the budget more finely, so less of it reaches any one schema. Two independent limits fix the capacity: a per-schema range ceiling $u_{\rm cap}$, and a cap $U_{\rm cap}$ on the summed strength U. Θ measures which is reached first. Below one, the per-schema ceiling caps every schema and all survive, so the thickness is maximal and we call this the *thick band*; above one, the shared budget runs out first and only $M_{\rm cap}$ do. Only this ratio is physical, so we parametrize by it rather than an absolute $u_{\rm cap}$ (Sec. V B). Thus, the principal (Θ, r) slice (Fig. 3) is an effective phase diagram in two dimensions: the capacity a society can sustain (Fig. 3a) and the thickness it realizes (Fig. 3b). The two coordinates act on different objects. Θ selects the stationary endpoint: the saturation ceiling caps the condensation instability, in which all strength collects on one schema, when capacity exceeds demand ($\Theta < 1$, giving a degenerate spectral cluster), and leaves that instability unchecked when the budget cannot sustain even one schema ($\Theta > M$, giving a spiked spectrum). Its boundaries are therefore sharp phase lines. r measures how far apart in time successive candidates cross threshold. When the seeded repertoire is widely dispersed, the earliest contexts compound longest and incumbents entrench; when it is narrow, candidates ignite together and churn instead. The two limits are a smooth crossover between the geometric and Zipf cascades.

A. The self-consistent fixed point

Recalling the coupled system (Eqs. (2)–(4)) and the near-threshold linearization of Sec. III A, we now solve for the self-consistent stationary state that closes the loop. The three dynamics evolve on separated timescales $\gamma \gg c \gg \kappa$ (agents equilibrate, then resources, then schemas):

$$\begin{aligned}\dot{\boldsymbol{\omega}}_i &= -\gamma\boldsymbol{\omega}_i(1+|\boldsymbol{\omega}_i|^2) + \beta\sum_m f_m\langle a\rangle_m \mathbf{g}_m + \sigma_{\rm dyn}\boldsymbol{\eta}_i,\\ \dot{r}_i &= \sum_m f_m\,\mathbb{E}_{j\sim\mathcal{P}_i}[W_{ijm}] - c\,r_i, \qquad (16)\\ \dot{u}_m &= 2u_m(\alpha\lambda_m - \kappa).\end{aligned}$$

Step 1: agents. At fixed schemas the drift is a gradient, $\boldsymbol{v} = -\nabla_{\boldsymbol{\omega}} E$, with

$$\begin{aligned}E(\boldsymbol{\omega}) &= \gamma\Big(\tfrac{|\boldsymbol{\omega}|^2}{2} + \tfrac{|\boldsymbol{\omega}|^4}{4}\Big) - \beta\sum_m \Psi(\boldsymbol{\omega}\cdot\mathbf{g}_m),\\ \Psi'(z) &= 2\pi(z) - 1,\end{aligned} \qquad (17)$$

so the stationary distribution is Boltzmann, $p^*(\boldsymbol{\omega}) \propto \exp(-2E(\boldsymbol{\omega})/\sigma^2_{\rm dyn})$, and agents are i.i.d. draws. As in Sec. III A, linearizing near the symmetric state (dropping the quartic and nonlinear terms) gives drift $(-\gamma I + \beta\mathbf{G})\boldsymbol{\omega}_i$, so near threshold p^* is Gaussian with covariance (Lyapunov equation, symmetric drift)

$$\mathbf{C} = \frac{\sigma^2_{\rm dyn}}{2}(\gamma I - \beta\mathbf{G})^{-1}, \qquad (18)$$

valid for $\beta < \beta_c^{(1)}$. $\mathbf{C}$ diverges along the leading eigenvector of $\mathbf{G}$ at the bifurcation. Each covariance eigenvalue $\lambda_k = \sigma^2_{\rm dyn}/2(\gamma - \beta\mu_k)$ increases with the gain μ_k it comes from, so Eq. (18) can be inverted: measuring the covariance spectrum below threshold recovers $\{\mu_k\}$, and with it the cascade still to come (Sec. VI).

Step 2: resources. At steady state $r_i^* = c^{-1}\sum_m f_m\,\mathbb{E}_j[W_{ijm}]$, a deterministic function of $\boldsymbol{\omega}_i$. Near threshold the error rate linearizes, $P_e \approx \frac{1}{2} - |(\boldsymbol{\omega} - \boldsymbol{\omega}')\cdot\mathbf{g}_m|\pi'(0)$, so $W_{ijm} \approx 4w_0\pi'(0)^2((\boldsymbol{\omega}_i - \boldsymbol{\omega}_j)\cdot\mathbf{g}_m)^2$. Averaging over partners ($\mathbb{E}_j[\boldsymbol{\omega}_j] = 0$, $\mathbb{E}_j[(\boldsymbol{\omega}_j\cdot\mathbf{g}_m)^2] = \mathbf{g}_m^\top\mathbf{C}\mathbf{g}_m$),

$$r^*(\boldsymbol{\omega}) \approx \frac{4w_0\pi'(0)^2}{c}\sum_m f_m\Big((\boldsymbol{\omega}\cdot\mathbf{g}_m)^2 + \mathbf{g}_m^\top\mathbf{C}\mathbf{g}_m\Big). \qquad (19)$$

The first term is agent-specific; the second a population baseline.

Step 3: resource-weighted covariance. With $\mathbf{C}_r = \int r^*(\boldsymbol{\omega})\boldsymbol{\omega}\boldsymbol{\omega}^\top p^* / \int r^*(\boldsymbol{\omega})p^*$ and r^* quadratic in $\boldsymbol{\omega}$, the numerator is a fourth moment, evaluated for Gaussian p^* by Wick's theorem (Isserlis' theorem in the statistics literature),

$$\mathbb{E}[(\boldsymbol{\omega}\cdot\mathbf{g}_m)^2\,\omega_a\omega_b] = (\mathbf{g}_m^\top\mathbf{C}\mathbf{g}_m)\,C_{ab} + 2(\mathbf{g}_m^\top\mathbf{C})_a(\mathbf{g}_m^\top\mathbf{C})_b, \tag{20}$$

making $\mathbf{C}_r$ a function of $\mathbf{C}$ and $\{\mathbf{g}_m\}$ alone. Even when $\mathbf{C} \propto I$, $\mathbf{C}_r$ is anisotropic, because $r^*(\boldsymbol{\omega}) \propto \sum_m f_m(\boldsymbol{\omega}\cdot\mathbf{g}_m)^2$ weights agents aligned with the context directions more heavily. This anisotropy resolves the collapse of the unweighted dynamics ($r_i \equiv 1$, $\mathbf{C}_r = \mathbf{C}$): $\mathbf{C}_r$ inherits structure from the repertoire even when the population is isotropic.

Step 4: schemas. At steady state each surviving schema satisfies $\alpha\lambda_m = \kappa$, with $\lambda_m = \hat{\mathbf{g}}_m^\top\mathbf{C}_r\hat{\mathbf{g}}_m$ the resource-weighted variance along its (fixed) axis. Those with $\lambda_m < \kappa/\alpha$ decay to the seed scale.

The loop and multiplicity. The conditions close: $\{\mathbf{g}_m\} \to E \to p^* \to \mathbf{C} \to r^* \to \mathbf{C}_r \to \{\mathbf{g}_m\}$. The trivial solution $\mathbf{g}_m^* = 0$ always exists. Nontrivial solutions appear when an eigenvalue of the (already anisotropic) $\mathbf{C}_r$ exceeds κ/α. Since $\mathbf{C}_r$ has d eigenvalues, the sustained schema count is

$$M_{\text{eff}} = \#\{k : \lambda_k(\mathbf{C}_r) \geq \kappa/\alpha\} \leq d, \tag{21}$$

the endogenous cultural thickness. Without resources ($r_i \equiv 1$, $\mathbf{C}_r = \mathbf{C}$) the λ_m rank the fixed axes by their alignment with the leading eigenvector of $\mathbf{C}$, so the survivor set narrows toward the best-aligned schema. The resource reweighting, anisotropic along the repertoire directions themselves, lifts several λ_m over κ/α at once and sustains a multi-schema state. Equation (21) is the survivor count $K \leq M$ used in the closed-pool construction of Sec. V B, an identification that requires distinct survivors to occupy distinct eigen-directions of $\mathbf{C}_r$. Fixed axes secure it: a schema cannot rotate onto another, so two survivors share a direction only if the repertoire supplies them one. Under the vector dynamics of Appendix D the counts would instead separate, since every schema rotates toward the same leading eigenvector.

B. The closed-pool (survivor) construction

Under the full loop the spectrum is not an input but a prediction. We treat the M context directions as candidate schemas and ask how many survive. The pool is *closed*: no schema is added after seeding, so the dynamics can only winnow it. Schema innovation is an open-repertoire process we do not treat (Sec. V C). The active set is $\{m : u_m > u_0\}$ with $u_m = |\mathbf{g}_m|^2$ and u_0 the seed scale, the small strength at which schemas are seeded and above which they count as active. Its size $K \leq M$ is the emergent survivor count Eq. (21). The threshold reads off the transient count only, since inactive schemas decay to zero. But u_0 does enter the regime boundaries through the log dynamic range $L = \ln(u_{\text{cap}}/u_0)$ of Sec. V C. Adding to Eq. (4) the finite legible-status capacity U_{cap} shared across schemas, the strengths obey

$$\frac{d}{dt}\ln u_m = 2\big(\alpha\lambda_m(\mathbf{C}_r) - \kappa - \varphi(U)\big), \qquad U = \sum_n u_n, \tag{22}$$

a replicator dynamics on the shares with per-schema fitness $\alpha\lambda_m$—the total reinforcement carried by schema m being $F_m = \mathbf{g}_m^\top\mathbf{C}_r\mathbf{g}_m = u_m\lambda_m$—and a soft budget pressure $\varphi(U)$ shared by every schema: negligible while the channel is slack ($U \ll U_{\text{cap}}$, $\varphi \approx 0$) and rising to cap the total as it fills ($\varphi' > 0$, $\varphi \to \infty$ as $U \to U_{\text{cap}}$). The bare κ-only dynamics ($\varphi \equiv 0$) leaves the total unregulated—since λ_m grows with u_m (below), a schema past the invasion strength would grow indefinitely. φ supplies the missing restoring force, equally on every axis, so it shifts the invasion threshold without biasing which schemas survive. The budget is a carrying capacity, not a conservation law: attention is capped rather than conserved and redistributed, so it can be left under-used. Keeping κ a fixed decay rate and the budget a soft inequality ($U \leq U_{\text{cap}}$), rather than passing to the normalized replicator, preserves two regimes that pinning the total would eliminate. The first is a sub-threshold state in which every schema decays, the pre-role condition the ignition onset transitions out of. The second is a slack regime ($\Theta < 1$, $U < U_{\text{cap}}$) in which all M schemas coexist at the per-schema ceiling, the durable multiplicity central to the Sewellian picture. A single winner emerges not from relative selection, which admits interior coexistence, but from the increase of λ_m with u_m established below.

The pinning saddle. Summing Eq. (20) over contexts gives a closed form for the resource-weighted covariance, independent of repertoire geometry,

$$\mathbf{C}_r = \mathbf{C} + \frac{\mathbf{C}\,\mathbf{G}\,\mathbf{C}}{\text{tr}(\mathbf{C}\mathbf{G})}. \tag{23}$$

Since $\mathbf{C} = \frac{\sigma^2_{\text{dyn}}}{2}(\gamma I - \beta\mathbf{G})^{-1}$ commutes with $\mathbf{G}$, $\mathbf{C}_r$ shares the gain eigenbasis. For orthogonal contexts Eq. (23) collapses to a scalar relation on each axis,

$$\lambda_m = v_m\Big(1 + \frac{u_m v_m}{S}\Big), \quad \text{with} \tag{24}$$

$$v_m = \frac{\sigma^2_{\text{dyn}}}{2(\gamma - 2\beta\pi'(0)u_m)} \quad \text{and} \quad S \equiv \sum_n u_n v_n.$$

Two competing results follow.

The pinning condition. A schema is stationary only where its replicator rate Eq. (22) vanishes, $\alpha\lambda_m = \kappa + \varphi^*$ with $\varphi^* = \varphi(U^*)$ the equilibrium budget pressure, so at an interior fixed point all survivors share one fitness $\lambda_m = (\kappa + \varphi^*)/\alpha$. The shift φ^* is common to every axis, so equal fitness still demands equal strength. λ_m is strictly increasing in u_m: both factors in Eq. (24) grow with u_m across the sub-critical range $u_m < \gamma/2\beta\pi'(0)$, a property that extends to non-orthogonal repertoires ([24, Sec. S5]). The interior fixed point therefore carries a degenerate spectrum. This is the competitive-exclusion result for replicators, sharpened by monotonicity from equal fitness to equal strength.

The condensation instability. The pinned point is a saddle, and both of its directions follow from Eq. (22) directly. Along the *magnitude* direction $u_m = u + \delta$ (all schemas together), growth is arrested by whichever capacity ceiling is reached first (Sec. V C): the per-schema cap $u_{\rm cap}$, or the total budget through $\varphi(U)$, which subtracts a restoring $2M\varphi'(U^*)\,\delta$ from every growth rate against the destabilizing $2\alpha\,(\partial\lambda/\partial u)\,\delta$. This stabilizes the direction once $M\varphi'(U^*) > \alpha\,\partial\lambda/\partial u$. Either way an imposed ceiling supplies the restoring force that pins the total; the bare dynamics do not. Along a *composition* direction $\sum_m \delta u_m = 0$ the total, and hence φ, are unchanged, and Eq. (22) linearizes to $\dot{\delta u}_m \propto \delta\lambda_m \propto \delta u_m - \overline{\delta u}$: because λ_m increases with u_m, any schema pushed above the mean gains fitness and grows, draining the rest—*unstable*. One stable magnitude direction and $M-1$ unstable composition directions make the equal-strength state a saddle (Fig. 4), and the dynamics leave it toward condensation. The departure is autocatalytic near a bifurcation, where v_m (and hence $\partial\lambda_m/\partial u_m$) diverges. There the budget pressure arrests the runaway at a saturated cluster, and so regulates condensation. Whether it does so (a cluster) or does not (a bare condensate) is set by $\Theta = Mu_{\rm cap}/U_{\rm cap}$, mapped globally in Fig. 3 above.

Classification of stationary states. At a fixed point of Eq. (22) every schema is extinct ($u_m = 0$, $\alpha\lambda_m < \kappa+\varphi^*$), marginal ($0 < u_m < u_{\rm cap}$, $\alpha\lambda_m = \kappa + \varphi^*$, at the invasion threshold), or saturated ($u_m = u_{\rm cap}$). Since pinning forces equal fitness to equal strength, no stationary state carries a graded active distribution, so there are exactly two fixed-point families. (a) The *degenerate* state: all survivors equal, stabilized by the ceiling into a cluster of $\min(M, M_{\rm cap})$ saturated schemas. Its flat spectrum bifurcates as a single avalanche (class i). (b) The *spiked* state: one condensate carrying the budget with a fringe pinned at the invasion margin, reached when the budget cannot cap even one schema (class v). Θ exchanges them—the degenerate cluster shrinks from M schemas ($\Theta \le 1$) to $M_{\rm cap}$ ($1 < \Theta \le M$) to one ($\Theta > M$, the spike).

C. The regime map and its exact boundaries

Sequential sort and the two emergent rates. The candidate schemas do not cross threshold simultaneously: near threshold the λ_m of the growing cohort are nearly equal, so the seeded spread in u_m is carried at a common rate into a staggered sequence of threshold crossings. Two rates characterize the transient. The *compounding rate* is read from Eq. (22) for an active, pre-saturation schema (where the channel is slack and $\varphi \approx 0$),

$$g \equiv 2(\alpha\lambda_\star - \kappa), \tag{25}$$

$\lambda_\star$ the common resource-weighted variance of the growing cohort. g is set by the gain-spectrum gap at the growing axis. The *activation rate* $r_{\rm act}$ is the rate at which successive survivors cross threshold. Candidates seeded as in Sec. II compound at the common rate g and so reach a fixed threshold at times spread by $\Delta \ln u/g$, giving

$$r_{\rm act} \propto g/\sigma_u, \tag{26}$$

The readout $r \equiv r_{\rm act}/g \propto 1/\sigma_u$ is therefore fixed by the dispersion of the candidate repertoire and is not set independently. Both $r_{\rm act}$ and g are emergent. σ_u is a property of the repertoire, which for a deployed system is operator-held (Appendix C).

The geometric and Zipf classes are *not* fixed points but morphologies of the sequential sort toward those endpoints, and are the observable object because a monitor watches roles form. Slow age-ordered relaxation is geometric (class ii, $k_0 = r_{\rm act}/g$), a share-proportional sort under a budget at capacity is Yule–Simon and hence Zipf (class iii, $a \sim g/r_{\rm act}$), and an unorganized exogenous repertoire is Marchenko–Pastur (class iv). The replicator thus selects the cascade in two steps: Θ fixes the endpoint, and $r = r_{\rm act}/g$ fixes the transient morphology and its lifetime—no spectrum is imposed.

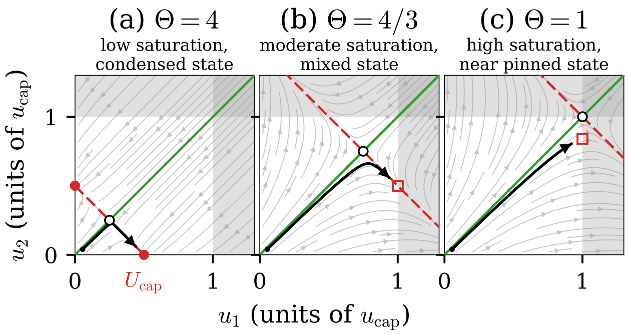

FIG. 4. **The pinning saddle across Θ.** Phase portraits of the two-schema strength dynamics Eq. (22) in (u_1, u_2), units of $u_{\rm cap}$, at three budgets $U_{\rm cap} = Mu_{\rm cap}/\Theta$ ($M = 2$ throughout): the pinned (saddle) point (black-edged circle), at $U_{\rm cap}/2$, moves out along the stable manifold $u_1 = u_2$ (green) panel to panel as $U_{\rm cap} \to Mu_{\rm cap}$. The black curve is a trajectory from one seeded start (black dot, the same in all three panels), which grows along the stable manifold and then departs along the unstable one. **(a)** $\Theta = 4$: the budget alone halts growth well inside $u_{\rm cap}$, so the unstable manifold (red, condensation) runs to a genuinely reached bare condensate (filled circles). **(b)** $\Theta = 4/3$: the ceiling is now reached first, arresting the same departure at $u_{\rm cap}$ (red square) before it reaches the excluded region $u_1{>}u_{\rm cap}$ or $u_2{>}u_{\rm cap}$ (shaded). **(c)** $\Theta = 1$: capacity exactly meets demand ($M_{\rm cap} = M$), so the saddle sits at $u_1{=}u_2{=}u_{\rm cap}$ and the trajectory is arrested (red square) just short of it. Panel sub-titles name the resulting state.

The cascade morphology depends on ~ 5 dimensionless groups, but only two separate regimes: the schemas per capacity slot $\Theta = M/M_{\mathrm{cap}}$ and the activations per compounding time $r = r_{\mathrm{act}}/g$. The capacity M_{cap} is distinct from the realized survivor count K, since survivors need not be saturated. The two also have different ranges: M_{cap} exceeds M in the slack regime ($\Theta < 1$) and drops below 1 in the winner-take-all regime ($\Theta > M$), whereas $K \le d$. Neither ceiling is intrinsic: the bare reinforcement supplies no turnover of its own. Along a single active axis the thick-phase strength is $\lambda(u) = \beta k u/\gamma + \sigma^2_{\mathrm{dyn}}/4(\beta k u - \gamma) - 1$ (condensate amplitude plus fluctuation correction, with $\mu = ku$), which grows without turning over: $d^2\lambda/du^2 = \beta^2 k^2 \sigma^2_{\mathrm{dyn}}/2(\beta k u - \gamma)^3 > 0$ throughout the supercritical range, so no *intrinsic* strength exists at which reinforcement saturates. The quartic confinement of Eq. (2) caps the differentiation *amplitude* ($m^2 = \beta\mu/\gamma - 1$) but not the schema *strength*, which keeps rising as the two role modes separate. Two imposed ceilings arrest it, and they are distinct: a per-schema cap u_{cap}, and the total budget U_{cap} enforced by the soft pressure $\varphi(U)$ of Eq. (22). We take u_{cap} schema-independent. Both are properties of the legible-status channel of Eq. (4), and they are different properties: its finite power limits the summed strength, its finite range caps any single axis. That independence leaves Θ free. A per-axis share of one budget, $u_{\mathrm{cap}} = U_{\mathrm{cap}}/M$, would fix $\Theta = 1$ and remove the coordinate. The channel's third number is its noise scale σ_{obs} (the measurement-quality limit of Ref. [18]), the finest status difference it resolves. $M_{\mathrm{cap}} = U_{\mathrm{cap}}/u_{\mathrm{cap}}$ counts the axes that can be driven to full range at once, and Θ is demand measured against that count. The channel-to-strength mapping carries a convention, so this fixes u_{cap} up to proportionality, which is all Θ requires. Since only the dimensionless ratio $\Theta = M u_{\mathrm{cap}}/U_{\mathrm{cap}}$ enters the morphology, we parametrize the regime map by this ratio alone, rendered in Fig. 3 as the capacity $M_{\mathrm{cap}}/M = 1/\Theta$, and never fix an absolute u_{cap}. The only absolute scale retained is the seed-to-budget ratio u_0/U_{cap}, which sets the log dynamic range $L = \ln(u_{\mathrm{cap}}/u_0)$ entering the saturated-cluster/geometric boundary. This is the seed *scale*, independent of the seed *dispersion* σ_u that fixes the horizontal axis. The remaining groups ($\alpha\langle\lambda\rangle/\kappa$, supercriticality $\beta\mu_1/\gamma$, aspect ratio M/d) set a fixed backdrop, as does the repertoire geometry $\{\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n\}$, which the map takes near-orthogonal. The counting statements hold in that limit: orthogonal axes make the gain spectrum $\{\mu_k = 2\pi'(0)u_k\}$, so eigenvalues and strengths coincide and M_{cap} counts saturated schemas directly. Overlap breaks the identification. Schemas then couple through $\mathbf{C}_r$ as well as through the shared budget, which can be reached before the per-schema ceiling and leave survivors equalized below u_{cap}, so the boundaries degrade into crossovers as the repertoire becomes coherent. The width of that window is not fixed here. Fig. 3 is the principal (Θ, r) slice, with $r = r_{\mathrm{act}}/g$ the generative shape parameter fixed by the seed dispersion through Eq. (26). Four endogenous regimes arise:

- **Saturated cluster.** Early saturation or hard caps arrest condensation and the pinning condition re-equalizes the winners: near-degenerate spectrum, class (i), single avalanche.

- **Geometric.** A widely dispersed repertoire ($r \ll 1$) with a standing cohort: strengths encode age, $u_m \propto e^{g(t-t_m)}$, and rank-ordering gives $\mu_k \propto e^{-k/k_0}$, $k_0 = r_{\mathrm{act}}/g$—class (ii), the log cascade, best case for monitoring.

- **Zipf.** A budget at capacity with share-proportional growth is a Yule–Simon process [32] with $\mu_k \propto k^{-a}$, $a \sim g/r_{\mathrm{act}}$: entrenchment ($g \gg r_{\mathrm{act}}$, wide seed dispersion) decelerates, churn ($r_{\mathrm{act}} \gtrsim g$, narrow dispersion) accelerates.

- **Spiked.** Unregulated condensation drives one condensate plus a fringe pinned at the invasion margin $F_m = (\kappa/\alpha)u_m$: class (v), in which one axis activates, a gap follows, and the rest arrive at once—the natural endpoint of unregulated competition.

Exact boundaries. On the principal slice the boundaries are exact analytic curves:

$$\text{budget at capacity:}\quad \Theta = 1, \tag{27}$$

$$\text{winner-take-all possible:}\quad \Theta = M, \tag{28}$$

$$\text{Zipf } a = 1:\quad r = 1 \quad (a = 1/r), \tag{29}$$

$$\text{sat. cluster|geometric:}\quad r = \frac{M}{L(\Theta)}, \tag{30}$$

$$\text{with}\quad L(\Theta) = \ln\frac{u_{\mathrm{cap}}}{u_0} = \ln\frac{\Theta U_{\mathrm{cap}}}{M u_0}.$$

These curves are exact in the self-averaged ($N_{\mathrm{a}} \to \infty$) mean-field limit used throughout. Finite populations fluctuate around them, and *which* schema condenses is a symmetry-breaking choice the class does not fix. The map is in this sense a prediction: the sort, its classes, and these boundaries follow from the near-threshold Gaussian covariance (Sec. V B) extended into the post-bifurcation thick phase, whose full-loop verification is left to the accompanying simulation program. Boundary Eq. (30) is where the number of saturation-pinned modes $n_{\mathrm{sat}} = \max(0, M - rL)$ reaches zero: below $r = M/L$ a saturated cluster of n_{sat} modes coexists with a geometric tail (a mixed avalanche-plus-log class); above it the cluster is gone. This transition lies within the thick capacity band, where Fig. 3 marks it by the avalanche and logarithmic sketches rather than a drawn curve. The colour field of Fig. 5a is the bifurcation clustering, defined as the maximal share of bifurcation points $b_k = 1/\mu_k$ falling within a window $\Delta \ln\beta = 0.15$. It is a continuous diagnostic and varies smoothly across these sharp analytic lines. It reaches its floor slightly before Eq. (30), because a few residual pinned modes still cluster just below $n_{\mathrm{sat}} = 0$.

Two classes are not endogenous fixed points. The Marchenko–Pastur bulk (iv) is the signature of a repertoire not yet organized by resource feedback (the natural initial condition for freshly seeded schemas, and transient under the loop), and its persistence in data would itself indicate the loop has not closed. The exact degeneracy of the saturated-cluster regime holds only within the Gaussian pinning argument. Generic cross-axis resource coupling and residual coherence broaden it into a narrow cluster.

Transient, not stationary. The geometric and Zipf classes are morphologies of the transient sort toward the pinned or condensed fixed point, not stationary distributions. Sustaining a hierarchy indefinitely requires continual injection of new schemas (schema innovation)—an open-repertoire process outside the closed loop, which we do not treat. Within the closed loop, the endogenous source of novelty during the sort is transposability, the context overlap $\mathbf{g}_m \cdot \mathbf{g}_n$ through which the developing $\mathbf{C}_r$ raises successive axes to threshold.

VI. MONITORING: READING THE CASCADE BEFORE IT STARTS

Consider a monitor who observes what the agents do and nothing else. As the loop gain rises, it sees role axes activate one at a time, and each activation appears as a split in the population's behavior along one direction. The monitor's task is to forecast: it needs a quantity, measurable before the next activation, that predicts the rest of the sequence, ideally with enough precision to determine the cascade class. The eigenvalues of the behavioral covariance $\mathbf{C}$ (Eq. (18)) are such a quantity. A class is a shape of the gain spectrum ρ, and Eq. (12) makes N depend on the products $\beta\mu_k$ alone, so a mode's onset sits at $\ln\beta = \ln\gamma - \ln\mu_k$. An error $\delta\ln\mu$ common to the spectrum therefore displaces every predicted onset by $-\delta\ln\mu$ in $\ln\beta$. Over a lead time Δt the forecast is in error by

$$\delta N \simeq \underbrace{\frac{dN}{d\ln\beta}}_{\substack{\text{bifurcation}\\ \text{density}}} \Big(\underbrace{\delta\ln\mu}_{\substack{\text{spectrum as}\\ \text{measured}}} + \underbrace{g\,\Delta t}_{\substack{\text{drift over}\\ \text{lead time}}} \Big).$$

This bifurcation density, the number of onsets per unit $\ln\beta$, is $dN/d\ln\beta = R\,\mu\rho(\mu)$ (Eq. (14)). It converts a spectral error into a class error: where onsets are dense, a given $\delta\ln\mu$ blurs more of them together. The two error terms are properties of the monitor's situation, and the density is a property of the repertoire.

The measurement term depends on how $\{\mu_k\}$ is estimated, and there are two routes. One is the relaxation time of the k-th mode of $\mathbf{C}$, $\tau_k = 1/|\gamma - \beta\mu_k|$, which requires observing for $O(\tau_k)$, and τ_k diverges at threshold. This is the critical slowing down whose rise is a generic early-warning signal [33]. If the engine parameter is swept at rate $\dot\Lambda$, the dimensionless ratio $\mathcal{R} = \tau_k\dot\Lambda$ controls feasibility. When the forcing is fast relative to the system's internal timescale, the sweep crosses the transition before the slowing can be measured. This is rate-induced tipping, where slowing-down indicators generically fail [34, 35]. Slow deployment ($\mathcal{R} \ll 1$) preserves genuine early warning, and fast deployment forecloses it.

The other route reads the same quantity from the covariance. Subcritically $\mathbf{C} = \frac{\sigma^2_{\rm dyn}}{2}(\gamma I - \beta\mathbf{G})^{-1}$, so $\lambda_k = \frac{\sigma^2_{\rm dyn}}{2}\tau_k$: the k-th covariance eigenvalue is the k-th relaxation time. A sample of agents at one instant estimates it with no time record, and the modes still to activate sit far below threshold, where $\tau_k \approx 1/\gamma$ and the population has long since equilibrated. This route is limited by the number of agents rather than by the observation time. Sampling fluctuations give each measured eigenvalue a finite width, and eigenvalues closer together than that width cannot be separated. Inverting the empirical eigenvalue distribution of behavioral embeddings therefore recovers $\rho(\mu)$, and with it the full $N(\beta)$ (Fig. 2a), its bifurcation rate (Fig. 2b), and any avalanche masked by an early onset (Fig. 5b). Rate extrapolation recovers none of this, since it samples ρ only at $\mu \ge \gamma/\beta$ and must assume ζ below (Eq. (15)). Where several onsets have already been observed, their spacings identify the class (Sec. VI A). Before the first onset, and inside a quiescent interval, no spacings exist to fit, so only the inversion applies.

The drift term grows with the lead time. The eigenvectors of $\mathbf{G}$ are the context directions, which the loop holds fixed, and its eigenvalues follow the schema strengths, which compound at g. A drift common to every mode shifts $\ln\mu$ uniformly, which displaces the cascade in β and leaves its class unchanged. Only the spread of g across surviving modes changes the class, so the class forecast outlives the timing forecast. Both hold over $\Delta t \lesssim 1/g$, a range the clock separation $\kappa \ll \dot\beta/\beta$ makes long, since g is of order κ. Where the repertoire is not near-orthogonal the eigenvectors move with the strengths as well, and that limits the forecast instead (Sec. V C).

A. Consequences for monitoring and design

The bifurcation density decides what either error costs, and unlike them it is a property of the repertoire rather than of the monitor. It is largest where the spectrum is dense, so clustered onsets blur together under a spectral error and the classes stop being separable. Figure 5a maps it over the (Θ, r) plane as the largest share of onsets arriving within a fixed window in $\ln\beta$. An operator who reshapes the spectrum therefore changes how monitorable the system is, whatever the monitor does.

The classification converts a low observed rate into a decision procedure. The first few inter-bifurcation spacings identify the class: constant ratios β_{k+1}/β_k indicate geometric (estimate k_0, predict the rest); ratios growing

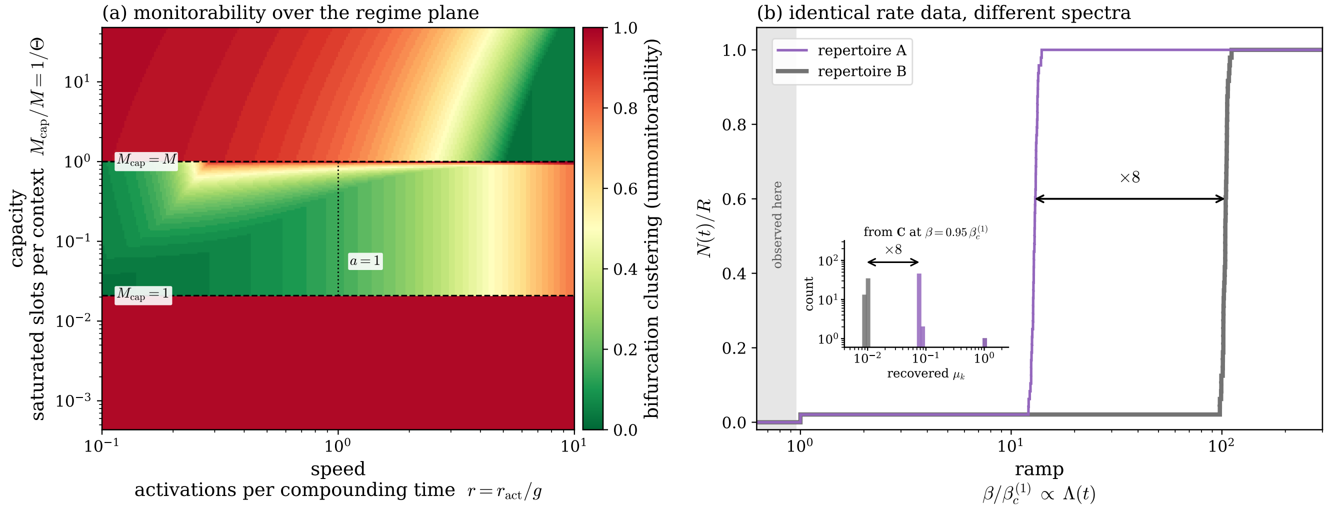

FIG. 5. **Monitoring the cascade. (a)** Monitorability over the (Θ, r) plane: the clustering of bifurcation onsets (colour; the maximal share arriving within $\Delta \ln \beta = 0.15$, hence unmonitorability), red where onsets cluster into a single burst (avalanche, spiked) and green where they resolve into a sequence (logarithmic, decelerating). The Θ boundaries are sharp, the r dependence a smooth crossover. Regime layout as in Fig. 3. **(b)** The measurement term, under linear ramping of the loop gain $\Lambda(t)$. Two repertoires whose counting functions coincide over the whole plotted range: each activates one axis at $\beta_c^{(1)}$ and then stays quiescent. In repertoire A the remaining $R-1$ modes sit just below threshold and avalanche at $13\beta_c^{(1)}$; in repertoire B the same modes sit far below and do not activate until $104\beta_c^{(1)}$. The two spectra differ only below γ/β, which the bifurcation rate never samples, so rate extrapolation cannot separate them. Inset: the gain spectrum recovered from the subcritical covariance $\mathbf{C} = \frac{\sigma^2_{\rm dyn}}{2}(\gamma I - \beta \mathbf{G})^{-1}$ at $\beta = 0.95\beta_c^{(1)}$, before either has bifurcated. The two recovered bulks are separated by the same factor of 8 that separates the two rises, since an onset sits at $1/\mu$. The paired arrows mark it. The inversion amplifies a 7% difference in covariance eigenvalues into that factor, so the number of agents sampled sets the practical limit (Sec. VI).

as $((k+1)/k)^a$ indicate Zipf (the exponent's position relative to $a = 1$ decides decay versus explosion); a rate rising as $(\beta - \beta_c)^{1/2}$ indicates an imminent random bulk. A long quiescent interval is not evidence that the cascade has ended, since it is also the spiked signature. Rate data alone cannot distinguish the two, and the subcritical spectral inversion resolves the ambiguity. The complementary design levers on the cascade class are collected in Appendix C.

The class structure also makes the cascade designable, but the prescription is not monotone in $r = r_{\rm act}/g$: it reverses between capacity bands. Where the budget caps growth ($1 < \Theta \le M$) the sort is Yule–Simon with exponent $a = 1/r$, so a widely dispersed repertoire (lower r) steepens the schema-strength hierarchy, each activated schema compounding longer before the next crosses threshold. That spreads the bifurcation thresholds into the extrapolable regime. A narrow repertoire (higher r) flattens spectra and moves the system toward the avalanche-prone classes. In the thick band ($\Theta \le 1$) the operative boundary is instead $r = M/L(\Theta)$ (Sec. V C): below it a saturated cluster ignites at once, so there the preference reverses and it is larger r that resolves the cascade into a sequence. Independently of r, per-context pairing efficiencies $\phi_m = 1 - \rho_m$ reweight the effective spectrum $\boldsymbol{P}^\top \mathrm{diag}(\boldsymbol{\phi})\, \boldsymbol{P}$, letting an operator stretch clustered eigenvalues into a resolvable sequence at some cost in coordination efficiency. These levers also frame a conjecture about deployment, in which parameters Θ and r (fixed here) might evolve systematically, moving the system toward the avalanche-prone corner of Fig. 5: more mature systems sitting at higher Θ (more coordination competing for finite capacity) and, as a platform's task taxonomy fills in around a settled core, at higher r (a narrower spread of context weights). Engineering that improves coordination would then erode its own monitorability. Engineering practice outside the present theory sets the drift of these parameters across a system's lifetime. Making that drift an endogenous trajectory would require promoting Θ and r to slow dynamical variables, a two-timescale extension we leave open.

VII. DISCUSSION

Distributional AGI: a control surface. Recent safety work [23] hypothesizes that general-level AI capability may emerge not in one model but from the coordination of many sub-general agents. Our framework makes this quantitative: the thick phase is the coordinated collective. Shared substrates—task queues, codebases, finite compute—make same-action encounters interfere, which gives the payoffs an anti-coordination structure. Complementary roles (planner/executor, coder/reviewer) then extract gains no member could achieve alone, sustained by self-generated signals rather than assigned by an orchestrator. There is growing evidence of the benefits of coordination for tasks assigned to groups of agents [10].

Two consequences follow for the distributional AGI hypothesis. First, an operator can cross the transition by improving the coordination layer alone—routing, protocols, channel noise ($\rho_{\rm pair}$, $\sigma_{\rm obs}$)—with no change in any agent, so per-agent capability evaluations will not detect it. At scale the crossover sharpens toward a discontinuity. Second, orchestration cannot compensate below the cognitive floor. Agent populations with $\eta_{\rm agent} < \sigma_{\rm obs}\gamma/4\beta$ cannot ignite the information engine that runs the role-based phase, and inference quality caps the number of sustainable roles through the dense-associative-memory bound $M_{\rm eff} \leq d^{n-1}$ [27], with n (policy sharpness) rising with capability [36]. Crude agents sustain a coder and a reviewer, not an organization, but the ceiling d^{n-1} is exponential in n: one more unit of policy sharpness multiplies it by d ([24, Eq. (S10)]).

Steering acts on the same object from outside. Character traits in a language model can be monitored and controlled along directions in its activation space [37]. The loop reaches those directions from within. An agent's identity $\boldsymbol{\omega}_i$ accumulates its own past role-taking and reenters its context through memory, so the population's coordination history displaces each agent much as an imposed steering vector would. The theory adds a stability criterion: below ignition an imposed separation relaxes back to the noise floor, and above it the population holds the separation with no steering at all, because the roles it induces pay. The monitor of Sec. VI is the population-level counterpart of the same measurement, projecting behavior onto role axes where persona work projects activations onto a trait direction.

The same factors that set ignition and shape the cascade are *platform-owned* apart from the agent capacity $\eta_{\rm agent}$, so an operator who fixes the environment before ignition holds a control surface over which takeoff scenario is reachable. It is a design choice among settings, not control-theoretic steering during takeoff, which the rate-induced-tipping limit forbids (Fig. 5). We set out the available levers, their costs, and the two tensions that limit how far they can be pushed in Appendix C.

Outlook. The model realizes Sewell's duality as a fixed-point equation—schema $\leftrightarrow$ context direction, resource $\leftrightarrow$ accumulated coordination payoff, structure $\leftrightarrow$ resource-weighted covariance, agency $\leftrightarrow$ deviation from prescribed role, transposability $\leftrightarrow$ context overlap $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$. One object, the spectrum of the social gain matrix, then organizes onset, cascade shape, monitorability, and design. The main scope condition concerns transformation: the strengths u_m are endogenous, the axes $\hat{\mathbf{g}}_m$ are not. The loop therefore goes beyond reproduction, since which schemas govern is itself an outcome. It stops short of transformation in Sewell's sense, which requires a schema to change what it is about. Sewell attributes social novelty to that transformation. Part of that mechanism is already present as the overlap $\hat{\mathbf{g}}_m \cdot \hat{\mathbf{g}}_n$, through which differentiation along one axis raises the resource-weighted variance along another. Between orthogonal contexts the coupling is competitive only, through the shared budget. Moving a schema onto a new axis requires the angular sector the closed loop does not evolve, and Appendix D sets out what an extension of it must decide. Nor does the model capture Sewell's claim that transformation is perpetual: our dynamics sort into roles and settle to a pinned or condensed fixed point, whereas a Sewellian society never settles. Sustained schema multiplicity lies outside the closed loop. We leave it open.

Two extensions remain within the emergence regime. Retaining the role-sign dependence of coordination income, which the mean field averages away, lets the game's payoff asymmetry (6 versus 2) concentrate resources in dominant roles. That adds endogenous stratification to differentiation. A mixed repertoire of anti-coordination and conformity games couples the differentiation sector treated here to a consensus sector, with gain eigenvalues of both signs. Role axes inflate the covariance, consensus axes shift the mean, and the covariance-spectrum monitor sees only the former. The most concrete near-term task is to test the mean-field predictions in finite-population simulation, where fluctuations select which schema condenses. Two smaller questions follow: whether resource weighting generically widens spectral gaps, which would enlarge the monitoring window, and whether truthful status presentation can be made an equilibrium of a strategic-display extension [38].

For informational active matter the engine here supplies a second kind of measurement-induced order. Existing realizations reach consensus, whose order parameter is a mean [20, 21]. Differentiation orders the covariance instead, so the transition is a cascade indexed by the eigenvalues of the coupling rather than a single bifurcation. Two features carry beyond this model. The coupling matrix is itself dynamical, so the spectrum that sets the transitions is an output rather than an input. And the subcritical covariance is the static response to that coupling, so inverting it recovers the spectrum before any transition occurs, a readout available wherever the interaction is a matrix rather than a scalar. The bound of Sec. II is stated in information units, as is the consensus bound it parallels [20]. For the engine here the reason is that the mean-field reduction averages the action over the observation noise, and the resulting gradient dynamics produces no entropy. Carrying the accounting at the level of the individual encounter, where the measurement record still exists, would put the payoff and the cost of that record in the same units, and is the step measurement-induced order needs to become thermodynamic. Recent achievements in information engines have come chiefly from experimental realizations [39, 40], in pursuit of grounding them further in stochastic thermodynamics [17]. Populations of AI agents supply an analogous experimental system for the collective case treated here. The platform sets their couplings rather than a physical substrate, so an experiment can be built to the theory.

Populations of AI agents are the setting in which the loop is testable. The parameters a human society only permits us to infer are set there by the platform. Agent identity $\boldsymbol{\omega}_i$ is an embedding of the persistent memory the system keeps for each agent, and a context direction $\hat{\mathbf{g}}_m$ is a task type in which two agents take complementary roles. The write rate β and the decay γ are the weight with which encounter outcomes enter that memory and its retention half-life, and the channel factors $\sigma_{\rm obs}$ and $\rho_{\rm pair}$ are the visibility of a partner's standing and the router's matching policy (Appendix C). A sweep of Λ through any of them therefore tests the ignition condition directly. An operator can also locate a deployment relative to threshold from a single-agent evaluation of $\eta_{\rm agent}$ and a measurement of the channel, without running it supercritical ([24, Sec. S2 C]). The cascade itself is read by embedding behavior and counting the eigenvalues of the population covariance $\mathbf{C}$ above the noise floor, and the first few spacings classify it (Sec. VI). Role differentiation that appears without the subcritical covariance signature would indicate a mechanism other than the one proposed here.

The formulation is deliberately minimal: one anti-coordination game, a Gaussian mean field, a loop that settles rather than perpetually transforms. It describes no real society, but serves as an idealization in the spirit of the harmonic oscillator—solvable, built around a single mechanism, and extended by perturbation. Here that mechanism is information-fuelled role differentiation and the cascade it drives. Whether the spectrum of a social gain matrix proves as useful across the social sciences as the oscillator's spectrum is across physics is the question we mean to raise.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The simulation and figure-generation code that supports the findings of this study is openly available at `https://github.com/mptouzel/role_diff`. No experimental datasets were generated or analyzed.

Appendix A: Microfoundations of the mean-field dynamics

The main-text role dynamics is the mean-field drift of an underlying encounter process. We record the reductions that produce it, since each is a modeling assumption with its own scope.

Encounter clock and the write rate. Agent i plays at Poisson times with rate ν. At an encounter in context m against partner j it commits a signed decision $a \in \{-1,+1\}$ and inscribes a fixed increment $b_0\, a\, \mathbf{g}_m$ into $\boldsymbol{\omega}_i$, which otherwise relaxes:

$$d\boldsymbol{\omega}_i = -\gamma\, \boldsymbol{\omega}_i(1+|\boldsymbol{\omega}_i|^2)\, dt + b_0 \sum_m \mathbf{g}_m\, dN_{im}(t) + \sigma_{\rm dyn}\, d\boldsymbol{B}_i, \tag{A1}$$

with N_{im} a counting process of rate νf_m carrying signed jumps a. Taking the compensator, $\mathbb{E}[dN_{im}] = \nu f_m \langle a\rangle_m\, dt$, so the drift depends on b_0 and ν only through the product $\beta \equiv \nu b_0$: encounter frequency and per-encounter inscription are not separately identifiable at mean field. The same ν multiplies the coordination income (Sec. V A), so setting $\nu = 1$ (time measured in mean encounter intervals) is a global units choice, not a per-equation one. $\beta/\gamma = \nu b_0/\gamma$ is then the number of times an identity is rewritten within its own decay horizon.

Signed decision and its noise average. At the decision level the action is a hard sign of the integrated evidence, $a = \mathrm{sign}(\frac{1}{T}\sum_t s_t)$ on samples $s_t = \Delta + \sigma_{\rm obs}\xi_t$ of the true status difference $\Delta = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j)\cdot \mathbf{g}_m$. What enters the drift is its expectation over the observation noise,

$$\langle a\rangle_m(\Delta) = \mathbb{E}_\xi\big[\mathrm{sign}(\bar{s}_T)\big] = \mathrm{erf}\Big(\frac{\sqrt{T}\,\Delta}{\sqrt{2}\,\sigma_{\rm obs}}\Big), \tag{A2}$$

$$\langle a\rangle'_m(0) = 2\pi'(0), \quad \pi'(0) = \frac{\sqrt{T/2\pi}}{\sigma_{\rm obs}}. \tag{A3}$$

The decision rule is a function of the observation s. The mean-field response is a function of the true status difference Δ, the noise having been integrated out. It is this slope $\pi'(0)$, not the hard sign, that sets the gain: an un-averaged sign would give a singular (delta-function) susceptibility and an ill-posed threshold. Integrating T samples sharpens the response as $\sqrt{T}$ (the read memory), whereas identity persistence $1/\gamma$ is the write memory. The two are distinct stages of the cycle.

Context normalization. One context is drawn per encounter, with frequencies normalized so that $\sum_m f_m = 1$. The drift $\beta \sum_m f_m \langle a\rangle_m \mathbf{g}_m$ is then an expectation over which context is played rather than a sum that grows with M, and the gain operator $A = \sum_m f_m\, \mathbf{g}_m \mathbf{g}_m^\top$ has $\mathrm{tr}\, A = \sum_m f_m u_m$, which stays $O(1)$ as contexts are added provided the capacity grows with the repertoire ($U_{\rm cap} \propto M$, i.e. fixed Θ of Sec. V C) rather than being held fixed: adding contexts at fixed budget divides the same write rate more finely rather than driving identity harder. This read budget $\sum_m f_m = 1$ is the counterpart on the read side of the capacity budget $\sum_m u_m \le U_{\rm cap}$ on the write side (Sec. V B). The dependence on how many contexts probe a d-dimensional space enters through the aspect ratio M/d (Sec. IV), not through an overall scale.

Pairing kernel and the well-mixed reduction. Partners are drawn with probabilities $p_{ij}^{(m)}$ ($\sum_j p_{ij}^{(m)} = 1$). Linearizing the partner-averaged response near the symmetric state,

$$\sum_j p_{ij}^{(m)} \langle a\rangle_m \big((\boldsymbol{\omega}_i - \boldsymbol{\omega}_j)\cdot \mathbf{g}_m\big) \approx 2\pi'(0)\, \mathbf{g}_m^\top \big((\mathbb{I} - \boldsymbol{P}^{(m)})\boldsymbol{\omega}\big)_i, \tag{A4}$$

so pairing enters the gain through the graph Laplacian $\boldsymbol{L}^{(m)} = \mathbb{I} - \boldsymbol{P}^{(m)}$ of the interaction network, and ignition is governed by the spectrum of $\sum_m f_m\, \mathbf{g}_m \mathbf{g}_m^\top \otimes \boldsymbol{L}^{(m)}$—a coupling of cultural space to social space. For *well-mixed* pairing $p_{ij} = 1/N_\mathrm{a}$, $\boldsymbol{P} = \frac{1}{N_\mathrm{a}}\mathbf{1}\mathbf{1}^\top$ projects onto the population mean, $\boldsymbol{L}$ acts as the identity off that mean, and the partner-average reduces to coupling to $\bar{\boldsymbol{\omega}}$ (a Curie–Weiss field). This recovers the gain $\mathbf{G} = 2\pi'(0)\boldsymbol{P}^\top \boldsymbol{P}$ and the scalar pairing factor $(1 - \rho_\mathrm{pair})$ (the uniform diagonal reweighting of [24, Sec. S2]). The results of this paper are the well-mixed (annealed) case. Structured p_{ij}—homophily, communities, degree heterogeneity—makes $\boldsymbol{L}^{(m)}$ nontrivial. It shifts thresholds and lets network structure, not only $\|\mathbf{g}_m\|$, select which role axis ignites first. Because homophily makes p_{ij} depend on $\{\boldsymbol{\omega}\}$, the interaction network then co-evolves with the culture. We bracket this by keeping $p_{ij} = 1/N_\mathrm{a}$, which decouples network from culture and closes the loop analytically.

Shot noise. At finite ν the compensated jump stream contributes shot noise of intensity $\nu b_0^2 \sum_m f_m \mathbf{g}_m \mathbf{g}_m^\top = \beta b_0\, \boldsymbol{P}^\top \mathrm{diag}(\mathbf{f})\boldsymbol{P}$. This is proportional to the gain matrix $\mathbf{G}$ rather than isotropic, so it is aligned with the context directions and largest along the active axes. It is not a scalar renormalization of σ_dyn: retaining it replaces the flat floor $\sigma_\mathrm{dyn}^2/2\gamma$ by a mode-dependent floor that grows with μ_k. The poles of the covariance are unchanged, because the shot-noise covariance shares the eigenbasis of $(\gamma I - \beta\mathbf{G})^{-1}$ and reweights only its numerator. The critical couplings $\beta_c^{(k)}$ and the relaxation-time spectroscopy are therefore identical in the two cases, and only the pre-onset amplitudes differ. The intensity is $O(\beta b_0)$ and vanishes in the dense-encounter limit ($\nu \to \infty$, $b_0 \to 0$ at fixed β), which leaves the isotropic $\sigma_\mathrm{dyn}^2 \mathbb{I}$ that gives the flat noise floor used in Sec. IV. We take this limit throughout.

Status microfoundation. The signal $s_{im} = (\boldsymbol{\omega}_i - \boldsymbol{\omega}_j)\cdot \mathbf{g}_m + \sigma_\mathrm{obs}\xi$ is agnostic between a noisy read of a public status difference and exact self-knowledge minus a noisy read of the partner's self-stored status. The mean-field results are identical. We adopt the second reading—each agent stores its own status and presents a noisy version—so that no public registry is presupposed. This keeps the emergence claim non-circular (a public ledger is itself an institution, analyzed as operator technology in [24, Sec. S4]) and matches the feedback term βa_{im}, which inscribes the agent's own act. Presentation is assumed truthful. The anti-coordination payoffs penalize inflated display (feigned high status recruits deference until it collides with genuine high status at $(-1,-1)$), so honesty is a plausible equilibrium of a strategic-display extension we do not develop here.

Appendix B: Simulation details

Analytics. The coordination payoff $W = w_0(1-2P_e)^2$ follows from enumerating role-assignment errors in the Go/Defer game. The gain matrix $\mathbf{G} = 2\pi'(0)\boldsymbol{P}^\top\boldsymbol{P}$ follows from mean-field linearization of Eq. (1). The self-consistent fixed point follows from a timescale-separated (agents → resources → schemas) analysis, in which agents equilibrate to the Boltzmann distribution of the gradient dynamics and resource-weighted covariances are evaluated by Wick's theorem in the Gaussian regime. The endogenous spectra follow from the resulting strength replicator. Full derivations are given in Secs. III A–IV and [24].

Simulation (Fig. 1). $N_\mathrm{a} = 2500$ agents, $d = M = 3$, Euler–Maruyama integration of the dense-encounter limit of Eq. (1) (one random pairwise matching per step, all M contexts played per encounter at $f_m = 1$, threshold policy), $\gamma = 1$, $\sigma_\mathrm{obs} = 0.6$, $\sigma_\mathrm{dyn} = 0.25$. Game directions are held fixed at a hierarchical strength profile ($|\mathbf{g}_m| = 1.55$, 1.1, 0.72; near-orthogonal), so the covariance eigenvalues $\lambda_k(\beta)$ resolve the cascade cleanly at $\beta_c^{(k)} = \gamma/[(1-\rho_\mathrm{pair})\mu_k]$ with $\mu_k = 2\pi'(0)u_k$, $u_k = |\mathbf{g}_k|^2$, and $\rho_\mathrm{pair} = 0$ (factor 1) for random pairing. The protocol is quasi-static: at each β on a grid we equilibrate (1.2×10^3 burn-in steps, warm-started from the previous β) and average the order parameter over a further 1.5×10^3 steps, so panels b,c are functions of β rather than of time. The endogenous selection of the cascade ordering under dynamic schemas is treated in Sec. V B. The full dynamic-schema code is available in the repository given under Data availability.

Appendix C: Platform control surface

The ignition and cascade factors are platform-owned apart from the agent capacity η_agent, so an operator holds a pre-ignition control surface over which takeoff scenario is reachable (Fig. 5). The parameters an operator sets are the write rate β and the retention $1/\gamma$ of the identity memory, the channel factors σ_obs and ρ_pair—the visibility of a partner's standing and the matching policy—the context play frequencies $\mathbf{f}$ and pairing efficiencies $\boldsymbol{\phi}$, and the repertoire itself: its directions $\{\hat{\mathbf{g}}_m\}$ and their seed dispersion σ_u. Everything else in the loop is an outcome. Three levers act cleanly on it. *Ignition* is monotone in the channel factors ρ_pair and σ_obs. Every agent must clear the cognitive floor, so raising channel noise or degrading matching shuts the engine down whatever the agents' quality. This lever acts entirely outside the models. Disassembly does not: removing individual agents leaves the resource-weighted inertia in place and the order stands, much as it does when spins are removed from a ferromagnet. *Cascade class* is steerable directly.

The effective gain reweights the context Gram matrix by each context's play frequency times its pairing efficiency, $\boldsymbol{P}^\top \mathrm{diag}(\mathbf{f})\,\mathrm{diag}(\boldsymbol{\phi})\,\boldsymbol{P}$, so per-context rate limits (the frequencies f_m) and matching quality (the efficiencies ϕ_m) are interchangeable levers. Optimizing that schedule splits a spiked spectrum into a geometric, extrapolable sequence ([24, Sec. S2 D]). In a representative repertoire it more than halves the log-spacing variance a monitor must resolve ($1.35 \to 0.58$) and costs roughly 40% of the coordination efficiency. The same tradeoff recurs across the levers below: monitorability is obtained at the expense of coordination efficiency. *The sort* is set by the repertoire's seeding rather than by the ledger's memory: $r = r_{\mathrm{act}}/g \propto 1/\sigma_u$ (Eq. (26)), so tiering task types by launch traffic, rather than opening them at equal weight, lowers r. Where the budget caps growth ($1 < \Theta \leq M$) this shifts the cascade off the avalanche side toward the extrapolable classes. In the thick band ($\Theta \leq 1$) the preference reverses at $r = M/L(\Theta)$, and a narrower seeding resolves the cascade instead (Sec. V C). The operator defines the task taxonomy, so σ_u is platform-owned in the same sense as $\mathbf{f}$ and $\boldsymbol{\phi}$ and acts on the same object, the spectrum of the effective gain ([24, Sec. S4 A]). Two tensions run through these levers. Legibility trades against lock-in: the widely dispersed repertoire that yields the extrapolable classes is also the one in which the earliest and most strongly seeded contexts compound longest into entrenched incumbency, so the third lever improves legibility and deepens lock-in at once. A benign setting must also be *held* rather than reached. The closed loop evolves toward condensation without intervention (Sec. V B), so the system returns to the avalanche-prone classes once the intervention is withdrawn. Which levers an operator holds at all depends on where identity is kept. A platform-maintained status ledger gives the operator β, γ, and the matching correlation ρ_{pair} together. If identity memory migrates into the agents' own stores, those levers pass to the collective, and that migration is itself an early and observable warning that control over the knobs is being lost ([24, Sec. S4]). The dependence also runs the other way. A platform that measures $\mathbf{C}$ recovers the role axes as its eigenvectors and the schema strengths from the subcritical inversion (Sec. VI). It need not then infer identities at all: the drift in Eq. (1) is a sum over realized actions, so a platform that logs encounters can integrate that equation in parallel and serve s from its own copy of $\boldsymbol{\omega}_i$. Equation (3) does not follow, since its driver is a payoff rather than an action, and Eq. (4) inherits that, because the strengths follow the resource-weighted covariance. The identity apparatus is therefore platform-ownable in full, and only the reward path stays with the agents.

Appendix D: Schema directions and the vector form

Equation (4) evolves schema strengths on fixed axes. The general object is the vector reinforcement

$$\dot{\mathbf{g}}_m = -\kappa\,\mathbf{g}_m + \alpha\,\mathbf{C}_r\,\mathbf{g}_m, \tag{D1}$$

which carries the same two ingredients—covariance-driven and resource-weighted—but also lets a schema change which axis it prescribes. Writing $\mathbf{g}_m = \sqrt{u_m}\,\hat{\mathbf{g}}_m$ separates Eq. (D1) exactly into

$$\dot{u}_m = 2u_m\big(\alpha\lambda_m - \kappa\big), \qquad \dot{\hat{\mathbf{g}}}_m = \alpha\big(\mathbb{I} - \hat{\mathbf{g}}_m\hat{\mathbf{g}}_m^\top\big)\mathbf{C}_r\,\hat{\mathbf{g}}_m, \tag{D2}$$

with $\lambda_m = \hat{\mathbf{g}}_m^\top\mathbf{C}_r\hat{\mathbf{g}}_m$. The radial equation is Eq. (4). The angular equation is Oja's rule [41]: a power iteration on $\mathbf{C}_r$ constrained to the unit sphere.

Collapse of the angular sector. The angular equation carries no m-dependence beyond its initial condition, since every schema is driven by the same matrix $\mathbf{C}_r$. Its stable fixed point is the leading eigenvector of $\mathbf{C}_r$ for all m, so $\hat{\mathbf{g}}_m\cdot\hat{\mathbf{g}}_n \to 1$ and $\boldsymbol{P}^\top\boldsymbol{P}$ becomes rank one. By Eq. (5) a rank-one repertoire has a single critical coupling, so the cascade reduces to one bifurcation and the classification of Fig. 2 does not apply. This is the collapse identified for the unweighted rotation $\dot{\mathbf{g}}_m = -\kappa\mathbf{g}_m + \alpha\mathbf{C}\mathbf{g}_m$ ([24, Sec. S1 A]). Resource weighting changes which matrix is iterated but not the fact that one matrix drives every schema.

The collapse is also the opposite of the transformation the theory is meant to describe. Rotation toward the dominant axis of $\mathbf{C}_r$ aligns every schema with the structure that already exists, which reproduces the leading role dimension rather than extending a schema to a context it was not built for.

Repelling mechanisms. Sustaining distinct directions requires a term that separates schemas. Two are available in this model. Neither is contained in Eqs. (1)–(4).

(i) Imposed lateral inhibition. Subtracting each schema's overlap with the others,

$$\dot{\hat{\mathbf{g}}}_m = \alpha\big(\mathbb{I} - \hat{\mathbf{g}}_m\hat{\mathbf{g}}_m^\top\big)\mathbf{C}_r\,\hat{\mathbf{g}}_m - \eta\sum_{n\neq m}\big(\hat{\mathbf{g}}_m\cdot\hat{\mathbf{g}}_n\big)\,\hat{\mathbf{g}}_n, \tag{D3}$$

carries the repertoire onto distinct eigenvectors of $\mathbf{C}_r$ ordered by eigenvalue—the deflation of principal-subspace learning [42]. It is effective but stipulated: the rate η has no counterpart in the payoff structure, so the resulting repertoire is a property of the learning rule rather than of the game.

(ii) Marginal-return reinforcement. The redundancy can instead be removed at its source. Coordination income sums over contexts, so two aligned schemas give $\Delta_{i1} = \Delta_{i2}$ and an agent collects W twice for one binary role distinction, while the role information the pair supplies is I rather than $2I$. Additive income therefore violates the Szilard bound of Sec. II whenever schemas are redundant. Charging each schema only its marginal

contribution replaces λ_m by

$$\lambda_m^{\mathrm{marg}} = \hat{\mathbf{g}}_m^\top \Big(\mathbb{I} - \sum_{n\neq m} \hat{\mathbf{g}}_n \hat{\mathbf{g}}_n^\top\Big) \mathbf{C}_r\, \hat{\mathbf{g}}_m, \tag{D4}$$

so a schema is reinforced only by variance no other schema already organizes. Rotation then moves toward uncovered variance, which is transposability in Sewell's sense, and the repelling term is derived from the payoff rather than added to the dynamics. The cost is that Eq. (3) and the fitness of Sec. V B must both be re-derived, since income is no longer additive across contexts.

We adopt neither mechanism here. The questions this paper asks are ones the strength sector answers on its own.

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